

Enterprise Network Automation and Hybrid-Cloud Infrastructure as Code

Project ENT-HYBRID-NET — [GNS3](#) · [Ansible](#) · [Git](#) · [Terraform on AWS](#)

Item	Details
Unit	[Unit code and name]
Assessment	Group Project – Enterprise Network Design, Automation and Cloud IaC
Group	[Group name / number]
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1. Executive Summary

A multinational enterprise asked our consulting team to replace manual, device-by-device network administration with an automated, secure and repeatable hybrid-cloud platform. This report documents the complete solution we designed, built and tested.

On-premises network (Part A). We built a new GNS3 lab, ENT-HYBRID-NET, using a three-layer hierarchical design (core / distribution / access) that links Headquarters, a Branch Office and a Data Centre. It contains 4 Cisco 7200 routers, 6 Cisco EtherSwitch (NM-16ESW) switches, 6 Linux end-hosts/servers and a Debian-based Linux automation server (ANSIBLE-SRV). The network provides VLAN segmentation, router-on-a-stick inter-VLAN routing, multi-area OSPF with MD5 authentication and ABR summarisation, per-site DHCP, a central DNS server, NTP, remote Syslog, SSH-only management, four security ACLs and Internet access through PAT.

Automation (Part B). The configuration is described once, as YAML data in an Ansible inventory, and deployed by eight Ansible roles: backup, baseline hardening, switching, routing, services, ACLs, verification and compliance. Credentials are protected with Ansible Vault. Every change is version-controlled in Git, previewed with check/diff mode, applied idempotently, then automatically verified and audited.

Cloud (Part C). Terraform code provisions an AWS VPC with two public and two private subnets across two Availability Zones, an Internet Gateway, route tables, least-privilege security groups, a bastion host, a private Amazon Linux EC2 instance and an ED25519 SSH key pair. The code is validated and planned against the team's AWS account (25 resources), and the Git history shows it being built up one commit at a time.

All end-to-end tests passed: inter-site reachability, Internet access via NAT, DNS resolution, DHCP leasing, ACL enforcement, SSH-only management, NTP synchronisation and Syslog collection. The report closes with lessons learned and a roadmap: CI/CD pipelines, a source-of-truth (NetBox) and a site-to-site VPN to the AWS VPC.

2. Introduction

2.1 Scenario and objectives

The client is moving from traditional CLI-driven administration to an automated hybrid-cloud environment. Manual configuration causes inconsistent device settings, slow change delivery, configuration drift and poor audit trails. The engagement objectives were:

- Design and build a scalable, redundant, secure enterprise network for HQ, Branch and Data Centre (Part A).
- Automate configuration, backup, verification and compliance of every network device with Ansible (Part B).
- Provision secure AWS infrastructure with Terraform, stored and versioned in Git (Part C).
- Treat the network and cloud as code: declarative, reviewed, repeatable and testable.

2.2 Tools and platform

Component	Version / image	Role in project
GNS3 GUI + server	2.2.61 (local server + GNS3 VM on VirtualBox 7.2)	Network emulation, REST API used to build the topology
Routers	Cisco 7200 – IOS 15.0(1)M advenenterprise9	R1-HQ, R2-BR, R3-DC, R4-CORE
Switches	Cisco 2691 + NM-16ESW – IOS 12.4(23)	3 distribution + 3 access EtherSwitches
Linux hosts	Docker: alpine:3.20 (PCs), nginx:alpine (web server)	DHCP clients, test endpoints, web server
Automation server	Docker: python:3.12-slim (Debian 13) + ansible-core 2.17.14	Ansible control node, Syslog collector
Ansible collections	cisco.ios 11.6.0, ansible.netcommon 8.7.1, ansible.utils 6.1.1, ansible.posix 2.2.2	Network modules (network_cli over SSH/paramiko)
Terraform	1.16.2, hashicorp/aws 6.66, tls 4.4, local 2.9	AWS Infrastructure as Code
Git	2.54 + GitHub (private repository)	Version control for all code, configs and evidence
Host	Windows 11, Intel i9-14900HX, 32 GB RAM	Lab host

Table 1: Tools, versions and roles

2.3 Repository structure

```
Enterprise-Network-Automation/
├── gns3/          GNS3 REST client + topology builder (nodes, links, labels)
├── configs/       Day-0 device configurations (rendered, secrets as placeholders)
├── scripts/       config generator, console bootstrap, evidence tooling
├── docs/          IP plan, design justification, AWS architecture diagram
├── ansible/       ansible.cfg, inventory/ (source of truth), roles/, playbooks/, backups/,
reports/
├── terraform/     versions/variables/network/security/compute/outputs .tf
├── evidence/      raw command outputs, Ansible run logs, terraform plan
├── screenshots/   evidence screenshots used in this report
└── report/        this report and its generator
```

3. Enterprise Network Design (Part A)

3.1 Topology

Figure 1 shows the GNS3 topology, built from scratch with a Python script that calls the GNS3 REST API (gns3/build_topology.py). Because the whole lab is created as code, it can be rebuilt the same way every time. The layout follows the hierarchical model top to bottom: Internet edge and core, then the distribution layer (site routers and distribution switches), then the access layer (access switches and end hosts). Each site is drawn as its own coloured block, and every link is labelled with its subnet.

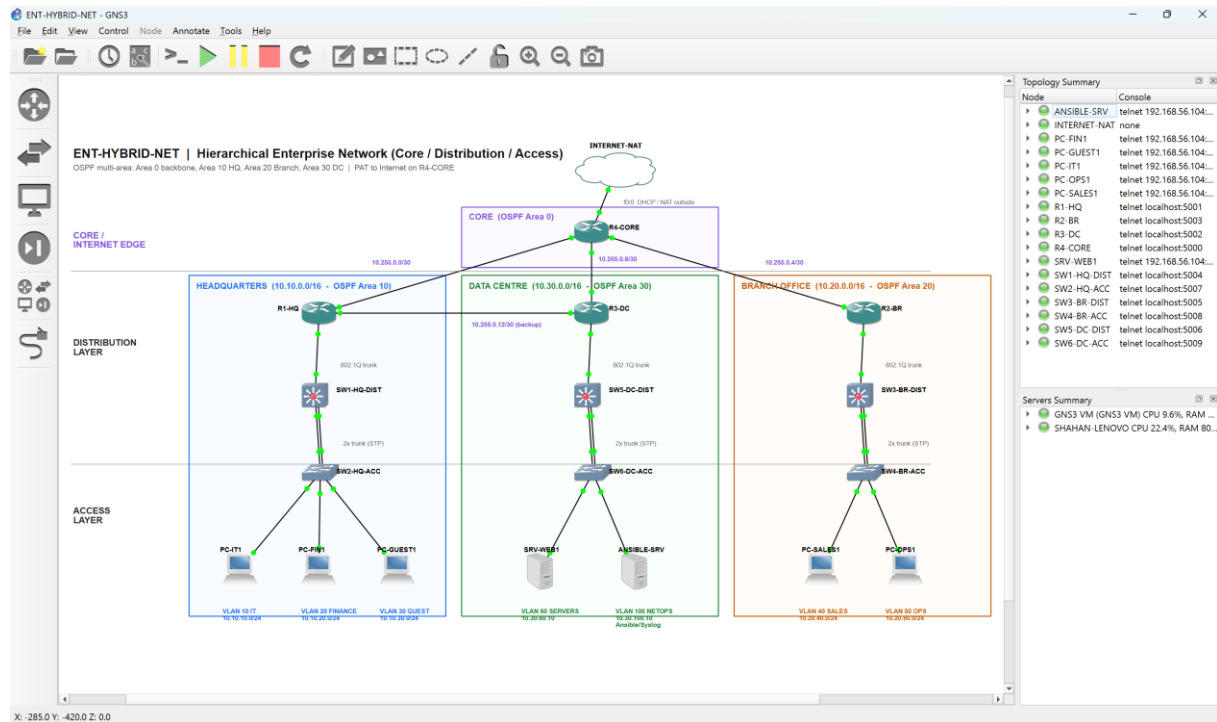


Figure 1: ENT-HYBRID-NET GNS3 topology – hierarchical core / distribution / access design

Layer	Devices	Functions
Core / Internet edge	R4-CORE, INTERNET-NAT	Backbone transit (OSPF area 0), PAT to Internet, NTP master, default-route origination
Distribution	R1-HQ, R2-BR, R3-DC (ABRs) SW1-HQ-DIST, SW3-BR-DIST, SW5-DC-DIST	Inter-VLAN routing, OSPF summarisation, ACL policy, DHCP, DNS (DC), STP root bridge
Access	SW2-HQ-ACC, SW4-BR-ACC, SW6-DC-ACC PCs, SRV-WEB1, ANSIBLE-SRV	Host attachment, VLAN assignment, PortFast, unused ports shut down

Table 2: Hierarchical layers and device roles

The minimum requirement of 4 routers, 6 switches and a Linux automation server is met: 4 × c7200 routers, 6 × c2691 EtherSwitch switches, and ANSIBLE-SRV (Debian 13 Linux container) in the Data Centre NETOPS VLAN.

3.2 IP addressing plan

Each site receives one summarisable /16 block, so each ABR advertises a single inter-area route into area 0. Loopbacks come from 10.0.0.0/24, and WAN links use /30s from 10.255.0.0/24. The AWS VPC

uses 172.20.0.0/16 so it never overlaps the on-premises ranges, which leaves a future site-to-site VPN possible.

Site	VLAN	Name	Subnet	Gateway	DHCP / notes
HQ	10	IT	10.10.10.0/24	10.10.10.1 (R1 Fa0/0.10)	Pool HQ-IT; may manage devices
HQ	20	FINANCE	10.10.20.0/24	10.10.20.1 (R1 Fa0/0.20)	Pool HQ-FINANCE; ACL FINANCE-PROTECT
HQ	30	GUEST	10.10.30.0/24	10.10.30.1 (R1 Fa0/0.30)	Pool HQ-GUEST; ACL GUEST-IN
HQ	99	MGMT	10.10.99.0/24	10.10.99.1	SW1 .11, SW2 .12 (static)
Branch	40	SALES	10.20.40.0/24	10.20.40.1 (R2 Fa0/0.40)	Pool BR-SALES
Branch	50	OPERATIONS	10.20.50.0/24	10.20.50.1 (R2 Fa0/0.50)	Pool BR-OPERATIONS
Branch	99	MGMT	10.20.99.0/24	10.20.99.1	SW3 .11, SW4 .12
DC	60	SERVERS	10.30.60.0/24	10.30.60.1 (R3 Fa0/0.60)	SRV-WEB1 10.30.60.10; ACL SERVERS-PROTECT
DC	100	NETOPS	10.30.100.0/24	10.30.100.1 (R3 Fa0/0.100)	ANSIBLE-SRV 10.30.100.10 (Syslog)
DC	99	MGMT	10.30.99.0/24	10.30.99.1	SW5 .11, SW6 .12

Table 3: VLAN and subnet allocation

Device	Interface	IP address	Connected to / purpose
R4-CORE	Lo0 / Fa0/0	10.0.0.4/32 / DHCP 192.168.42.x	Router-ID + NTP master / INTERNET-NAT (NAT outside)
R4-CORE	Fa1/0 · Fa1/1 · Fa2/0	10.255.0.1 · .5 · .9 /30	R1-HQ · R2-BR · R3-DC (NAT inside, area 0)
R1-HQ	Lo0 · Fa1/0 · Fa1/1	10.0.0.1 · 10.255.0.2 · 10.255.0.13	Router-ID · R4 · R3 backup link (cost 100)
R2-BR	Lo0 · Fa1/0	10.0.0.2 · 10.255.0.6	Router-ID · R4
R3-DC	Lo0 · Fa1/0 · Fa1/1	10.0.0.3 · 10.255.0.10 · 10.255.0.14	Router-ID + DNS server · R4 · R1 backup
R1/R2/R3	Fa0/0.<vlan>	.1 of each site VLAN	802.1Q trunk to the distribution switch
SW1 ... SW6	Vlan99	10.x0.99.11 (dist) / .12 (acc)	Management SVI, default gateway 10.x0.99.1
ANSIBLE-SRV	eth0	10.30.100.10/24	Automation + Syslog (ansible.corp.local)
SRV-WEB1	eth0	10.30.60.10/24	nginx web server (web.corp.local)
PCs	eth0	DHCP	PC-IT1, PC-FIN1, PC-GUEST1 (HQ) · PC-SALES1, PC-OPS1 (Branch)

Table 4: Device interface addressing

3.3 Configuration of required services

Requirement	Implementation	Devices
VLANs	VTP transparent; VLANs created in the NM-16ESW VLAN database; dual 802.1Q trunks dist↔access; STP root priority 4096 on distribution switch	SW1–SW6
Inter-VLAN routing	Router-on-a-stick: dot1Q sub-interfaces Fa0/0.<vlan> on each site router	R1, R2, R3
OSPF	Process 1, multi-area (0/10/20/30), ABR summarisation (area range), MD5 auth on area 0, passive-interface default, ref-bw 1000, default-information originate on R4	R1–R4
DHCP	Per-VLAN pools with excluded .1–.20, default-router, DNS 10.0.0.3, domain corp.local	R1, R2

Requirement	Implementation	Devices
SSH	SSH v2, RSA 2048, local user privilege 15 (secret), VTY transport input ssh + access-class MGMT-ACCESS	All
ACLs	MGMT-ACCESS (VTY), GUEST-IN, FINANCE-PROTECT, SERVERS-PROTECT, NAT-INSIDE	All / R1, R3, R4
NAT	PAT overload on R4 Fa0/0; NAT-INSIDE excludes internal-to-internal traffic	R4
DNS	ip dns server + ip host records for corp.local, forwards to 8.8.8.8; clients use 10.0.0.3	R3 (server), all (client)
NTP	R4 ntp master 3; all others ntp server 10.0.0.4; timezone AEST	All
Syslog	logging host 10.30.100.10, trap informational, source Loopback0/Vlan99, buffered 16 KB	All

Table 5: Required services and how they are configured

The full device configurations are in configs/*.cfg (Day-0, generated by scripts/gen_configs.py) and in ansible/backups/latest/*.cfg (the live running-configurations collected by Ansible). Key excerpts follow.

Excerpt – R1-HQ multi-area OSPF and DHCP (configs/R1-HQ.cfg)

```
router ospf 1
  router-id 10.0.0.1
  area 0 authentication message-digest
  area 10 range 10.10.0.0 255.255.0.0
  passive-interface default
  no passive-interface FastEthernet1/0
  no passive-interface FastEthernet1/1
  network 10.0.0.1 0.0.0.0 area 0
  network 10.255.0.0 0.0.0.3 area 0
  network 10.255.0.12 0.0.0.3 area 0
  network 10.10.10.0 0.0.0.255 area 10
  network 10.10.20.0 0.0.0.255 area 10
  network 10.10.30.0 0.0.0.255 area 10
  network 10.10.99.0 0.0.0.255 area 10
ip dhcp pool HQ-IT
  network 10.10.10.0 255.255.255.0
  default-router 10.10.10.1
  dns-server 10.0.0.3
  network 10.10.20.0 255.255.255.0
  dns-server 10.0.0.3
  network 10.10.30.0 255.255.255.0
  dns-server 10.0.0.3
```

Excerpt – R4-CORE NAT/PAT, NTP master, default route (configs/R4-CORE.cfg)

```
ntp master 3
description INTERNET (GNS3 NAT cloud) - PAT outside
ip nat outside
ip nat inside
ip nat inside
ip nat inside
! ===== NAT / PAT for all enterprise networks =====
ip access-list extended NAT-INSIDE
deny ip 10.0.0.0 0.255.255.255 10.0.0.0 0.255.255.255
permit ip 10.0.0.0 0.255.255.255 any
ip nat inside source list NAT-INSIDE interface FastEthernet0/0 overload
default-information originate always
```

Excerpt – SW2-HQ-ACC VLAN database and access ports (configs/SW2-HQ-ACC.cfg)

```
!EXEC-BEGIN
vlan database
vtp transparent
```

```

vlan 10 name IT
vlan 20 name FINANCE
vlan 30 name GUEST
vlan 99 name MGMT
apply
exit
!EXEC-END
switchport mode trunk
switchport mode trunk
interface FastEthernet1/5
switchport mode access
switchport access vlan 10
spanning-tree portfast
switchport mode access
switchport access vlan 20

```

3.4 Connectivity verification

The following screenshots come from live device consoles. Each window is titled with the device, the feature being verified and a timestamp.

```

Part A | R4-CORE | OSPF adjac  [localhost:5000]  2026-09-25 14:47:26
=====
R4-CORE#show ip ospf neighbor
Neighbor ID      Pri   State           Dead Time   Address        Interface
10.0.0.3          0    FULL/-          00:00:32    10.255.0.10    FastEthernet2/0
10.0.0.2          0    FULL/-          00:00:34    10.255.0.6     FastEthernet1/1
10.0.0.1          0    FULL/-          00:00:33    10.255.0.2     FastEthernet1/0
R4-CORE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is 192.168.42.1 to network 0.0.0.0

S* 0.0.0.0/0 [254/0] via 192.168.42.1
   10.0.0.0/8 is variably subnetted, 14 subnets, 3 masks
O   10.0.0.1/32 [110/11] via 10.255.0.2, 00:01:56, FastEthernet1/0
O   10.0.0.2/32 [110/11] via 10.255.0.6, 00:22:57, FastEthernet1/1
O   10.0.0.3/32 [110/11] via 10.255.0.10, 00:01:09, FastEthernet2/0
C   10.0.0.4/32 is directly connected, Loopback0
O IA 10.10.0.0/16 [110/20] via 10.255.0.2, 00:01:56, FastEthernet1/0
O IA 10.20.0.0/16 [110/20] via 10.255.0.6, 00:22:57, FastEthernet1/1
O IA 10.30.0.0/16 [110/20] via 10.255.0.10, 00:01:09, FastEthernet2/0
C   10.255.0.0/30 is directly connected, FastEthernet1/0
L   10.255.0.1/32 is directly connected, FastEthernet1/0
C   10.255.0.4/30 is directly connected, FastEthernet1/1
L   10.255.0.5/32 is directly connected, FastEthernet1/1
C   10.255.0.8/30 is directly connected, FastEthernet2/0
L   10.255.0.9/32 is directly connected, FastEthernet2/0
O   10.255.0.12/30 [110/110] via 10.255.0.10, 00:01:09, FastEthernet2/0
    [110/110] via 10.255.0.2, 00:01:56, FastEthernet1/0
    192.168.42.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.42.0/24 is directly connected, FastEthernet0/0
L   192.168.42.241/32 is directly connected, FastEthernet0/0
R4-CORE#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 2: R4-CORE – three FULL OSPF adjacencies, /16 inter-area summaries per site and the default route to the Internet

```

Part A | R1-HQ | Inter-VLAN su
=====
Part A | R1-HQ | Inter-VLAN sub-interfaces + DHCP bindings [localhost:5001] 2026-09-25 14:48:03
=====

R1-HQ#show ip interface brief
Interface              IP-Address      OK? Method Status      Protocol
FastEthernet0/0        unassigned      YES manual up          up
FastEthernet0/0.10     10.10.10.1      YES manual up          up
FastEthernet0/0.20     10.10.20.1      YES manual up          up
FastEthernet0/0.30     10.10.30.1      YES manual up          up
FastEthernet0/0.99     10.10.99.1      YES manual up          up
FastEthernet1/0        10.255.0.2      YES manual up          up
FastEthernet1/1        10.255.0.13     YES manual up          up
FastEthernet2/0        unassigned      YES unset administratively down down
FastEthernet2/1        unassigned      YES unset administratively down down
Loopback0              10.0.0.1        YES manual up          up

R1-HQ#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address             Client-ID/      Lease expiration    Type
                       Hardware address/
                       User name
10.10.10.23             0100.5079.6668.00 Sep 27 2026 12:30 AM Automatic
10.10.10.24             0102.425e.90d9.00 Sep 27 2026 12:31 AM Automatic
10.10.20.22             0100.5079.6668.01 Sep 27 2026 12:31 AM Automatic
10.10.20.23             0102.4224.0397.00 Sep 27 2026 12:32 AM Automatic
10.10.30.22             0102.428a.6ad1.00 Sep 27 2026 12:32 AM Automatic

R1-HQ#show ip dhcp pool | include Pool|Leased
Pool HQ-IT :
Leased addresses      : 2
Current index         IP address range      Leased addresses
Pool HQ-FINANCE :
Leased addresses      : 2
Current index         IP address range      Leased addresses
Pool HQ-GUEST :
Leased addresses      : 1
Current index         IP address range      Leased addresses
R1-HQ#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 3: R1-HQ – router-on-a-stick sub-interfaces and DHCP bindings for HQ clients

```

Part A | R1-HQ | Security ACLs
=====
Part A | R1-HQ | Security ACLs + OSPF interfaces (areas 0 and 10) [localhost:5001] 2026-09-25 14:48:25
=====

R1-HQ#show ip access-lists
Standard IP access list MGMT-ACCESS
10 permit 10.30.100.0 wildcard bits 0.0.0.255
20 permit 10.10.10.0 wildcard bits 0.0.0.255
30 deny any log
Extended IP access list FINANCE-PROTECT
10 permit ip 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255
20 permit ip 10.30.0.0 0.0.0.255 10.10.20.0 0.0.0.255
30 permit ip 10.0.0.0 0.0.0.255 10.10.20.0 0.0.0.255
40 deny ip 10.20.0.0 0.0.0.255 10.10.20.0 0.0.0.255 log
50 deny ip 10.10.30.0 0.0.0.255 10.10.20.0 0.0.0.255 log
60 permit ip any any
Extended IP access list GUEST-IN
10 permit udp any any eq bootps (24 matches)
20 permit udp 10.10.30.0 0.0.0.255 host 10.0.0.3 eq domain
30 permit icmp 10.10.30.0 0.0.0.255 host 10.10.30.1
40 deny ip 10.10.30.0 0.0.0.255 10.0.0.0 0.255.255.255 log
50 permit ip 10.10.30.0 0.0.0.255 any

R1-HQ#show ip ospf interface brief
Interface  PID  Area  IP Address/Mask  Cost  State  Nbrs  F/C
Lo0        1    0     10.0.0.1/32     1     LOOP  0/0
Fa1/1      1    0     10.255.0.13/30  100   P2P   1/1
Fa1/0      1    0     10.255.0.2/30   10    P2P   1/1
Fa0/0.99   1    10    10.10.99.1/24   10    DR    0/0
Fa0/0.30   1    10    10.10.30.1/24   10    DR    0/0
Fa0/0.20   1    10    10.10.20.1/24   10    DR    0/0
Fa0/0.10   1    10    10.10.10.1/24   10    DR    0/0
R1-HQ#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 4: R1-HQ – GUEST-IN, FINANCE-PROTECT and MGMT-ACCESS ACLs; OSPF interfaces in areas 0 and 10

```

Part A | R3-DC | DNS server (c)
MTID: 0      Metric: 30

LS age: 161
Options: (No TOS-capability, DC, Upward)
LS Type: Summary Links(Network)
Link State ID: 10.255.0.0 (summary Network Number)
Advertising Router: 10.0.0.3
LS Seq Number: 80000004
Checksum: 0x4BC6
Length: 28
Network Mask: /30
MTID: 0      Metric: 20

LS age: 161
Options: (No TOS-capability, DC, Upward)
LS Type: Summary Links(Network)
Link State ID: 10.255.0.4 (summary Network Number)
Advertising Router: 10.0.0.3
LS Seq Number: 80000001
Checksum: 0x29E7
Length: 28
Network Mask: /30
MTID: 0      Metric: 20

LS age: 1738
Options: (No TOS-capability, DC, Upward)
LS Type: Summary Links(Network)
Link State ID: 10.255.0.8 (summary Network Number)
Advertising Router: 10.0.0.3
LS Seq Number: 80000001
Checksum: 0x9C7A
Length: 28
Network Mask: /30
MTID: 0      Metric: 10

LS age: 1738
Options: (No TOS-capability, DC, Upward)
LS Type: Summary Links(Network)
Link State ID: 10.255.0.12 (summary Network Number)
Advertising Router: 10.0.0.3
LS Seq Number: 80000001
Checksum: 0xFBBC
Length: 28
Network Mask: /30
MTID: 0      Metric: 100

R3-DC#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 5: R3-DC – DNS server records for corp.local and OSPF database summary

```

Part A | R2-BR | Branch sub-in
=====
Part A | R2-BR | Branch sub-interfaces, DHCP bindings, OSPF [localhost:5003] 2026-09-25 15:03:13
=====

R2-BR#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0      unassigned      YES manual up           up
FastEthernet0/0.40    10.20.40.1      YES manual up           up
FastEthernet0/0.50    10.20.50.1      YES manual up           up
FastEthernet0/0.99    10.20.99.1      YES manual up           up
FastEthernet1/0       10.255.0.6      YES manual up           up
FastEthernet1/1       unassigned      YES unset administratively down down
FastEthernet2/0       unassigned      YES unset administratively down down
FastEthernet2/1       unassigned      YES unset administratively down down
Loopback0          10.0.0.2        YES manual up           up

R2-BR#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
Hardware address/
User name
10.20.40.21      0100.5079.6668.04  Sep 27 2026 12:29 AM    Automatic
10.20.40.22      0102.426f.5fca.00  Sep 27 2026 12:32 AM    Automatic
10.20.50.21      0100.5079.6668.05  Sep 27 2026 12:29 AM    Automatic
10.20.50.22      0102.42f4.ee4f.00  Sep 27 2026 12:32 AM    Automatic

R2-BR#show ip ospf neighbor
Neighbor ID      Pri   State           Dead Time   Address        Interface
10.0.0.4         0     FULL/ -         00:00:33    10.255.0.5     FastEthernet1/0
R2-BR#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 6: R2-BR – Branch sub-interfaces, DHCP bindings, OSPF adjacency and inter-area routes

```

Part A | SW1-HQ-DIST | VLAN
=====
Part A | SW1-HQ-DIST | VLAN database + 802.1Q trunks [localhost:5004] 2026-09-25 14:49:32
=====

SW1-HQ-DIST#show vlan-switch brief

VLAN Name                Status    Ports
-----
1    default                active    Fa1/3, Fa1/4, Fa1/5, Fa1/6, Fa1/7, Fa1/8, Fa1/9, Fa1/10, Fa1/11, Fa1/12, Fa1/13, Fa1/14, Fa1/15
10   IT                       active
20   FINANCE                 active
30   GUEST                   active
99   MGMT                     active
1002 fddi-default            active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active

SW1-HQ-DIST#show interfaces trunk

Port      Mode      Encapsulation  Status  Native vlan
Fa1/0     on        802.1q          trunking  1
Fa1/1     on        802.1q          trunking  1
Fa1/2     on        802.1q          trunking  1

Port      Vlans allowed on trunk
Fa1/0     1-1005
Fa1/1     1-1005
Fa1/2     1-1005

Port      Vlans allowed and active in management domain
Fa1/0     1,10,20,30,99
Fa1/1     1,10,20,30,99
Fa1/2     1,10,20,30,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa1/0     1,10,20,30,99
Fa1/1     1,10,20,30,99
Fa1/2     1,10,20,30,99

SW1-HQ-DIST#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 7: SW1-HQ-DIST – VLAN database and 802.1Q trunks

```

Part A | SW2-HQ-ACC | Spanning
=====
Port path cost 19, Port priority 128, Port Identifier 128.42.
Designated root has priority 4096, address c005.189c.0001
Designated bridge has priority 4096, address c005.189c.0001
Designated port id is 128.42, designated path cost 0
Timers: message age 1, forward delay 0, hold 0
Number of transitions to forwarding state: 1
BPDUs: sent 0, received 906

Port 43 (FastEthernet1/2) of VLAN10 is blocking
Port path cost 19, Port priority 128, Port Identifier 128.43.
Designated root has priority 4096, address c005.189c.0001
Designated bridge has priority 4096, address c005.189c.0001
Designated port id is 128.43, designated path cost 0
Timers: message age 1, forward delay 0, hold 0
Number of transitions to forwarding state: 0
BPDUs: sent 7, received 906

Port 46 (FastEthernet1/5) of VLAN10 is forwarding
Port path cost 19, Port priority 128, Port Identifier 128.46.
Designated root has priority 4096, address c005.189c.0001
Designated bridge has priority 32768, address c008.0c94.0001
Designated port id is 128.46, designated path cost 19
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
BPDUs: sent 910, received 0
The port is in the portfast mode

SW2-HQ-ACC#show interfaces status

Port  Name      Status      Vlan    Duplex  Speed  Type
Fa1/0  UNUSED      disabled    1       auto    auto   10/100BaseTX
Fa1/1  TRUNK to SW1-HQ-DI connected trunk  a-full  a-100  10/100BaseTX
Fa1/2  TRUNK to SW1-HQ-DI connected trunk  a-full  a-100  10/100BaseTX
Fa1/3  UNUSED      disabled    1       auto    auto   10/100BaseTX
Fa1/4  UNUSED      disabled    1       auto    auto   10/100BaseTX
Fa1/5  ACCESS PC-FIT1 (VLA connected 10    a-full  a-100  10/100BaseTX
Fa1/6  ACCESS PC-FIN1 (VL connected 20    a-full  a-100  10/100BaseTX
Fa1/7  ACCESS PC-GUEST1 ( connected 30    a-full  a-100  10/100BaseTX
Fa1/8  UNUSED      disabled    1       auto    auto   10/100BaseTX
Fa1/9  UNUSED      disabled    1       auto    auto   10/100BaseTX
Fa1/10 UNUSED      disabled    1       auto    auto   10/100BaseTX
Fa1/11 UNUSED      disabled    1       auto    auto   10/100BaseTX
Fa1/12 UNUSED      disabled    1       auto    auto   10/100BaseTX
Fa1/13 UNUSED      disabled    1       auto    auto   10/100BaseTX
Fa1/14 UNUSED      disabled    1       auto    auto   10/100BaseTX
Fa1/15 UNUSED      disabled    1       auto    auto   10/100BaseTX

SW2-HQ-ACC#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 8: SW2-HQ-ACC – spanning tree: redundant uplink Fa1/2 in BLK state (loop-free redundancy)

```

Part A | R1-HQ | SSH v2, NTP v2, NTP sync, Syslog, VTY security [localhost:5001] 2026-09-25 14:50:16
=====
R1-HQ#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAwEAAQc77yYhdE1B8nglgSBt62xOUxkVwXMIH8/
4EamwaurVd0D/3y4VLcVH8MkL6T8+NmZPnKqCqF3YmZMT2Rz019+TeNECv4h9FIS
16U6+Y1S8ad9TdLhDzD5c9V1QhLQ0F4nvtc+7Pk6eG066lWumFpsE9W+IF4+kXW
ao8xwRR1ZKwPailgnGeOzAqKJ3ahYz2m5sPhFXFOLyJUIKCsVMTjKAKHtPHV83N
pW3LZsT+183a1ZR9Y3Nqmg8VxpH4a9JQy9CcF4jqizM001XL6JldhTe8EGuUnyfr
w2SuLTFc16NC9Fdy3B96gtnn89Q8R4nhQoLyjTlG8Ze0vt03rNBX

R1-HQ#show ntp status
Clock is unsynchronized, stratum 16, no reference clock
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**24
reference time is 00000000.00000000 (10:00:00.000 AEST Mon Jan 1 1900)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 27.69 msec, peer dispersion is 0.00 msec
loopfilter state is 'FSET' (Drift set from file), drift is 0.000000000 s/s
system poll interval is 64, never updated.
R1-HQ#show ntp associations

 address      ref clock      st  when  poll reach delay offset disp
~10.0.0.4      127.127.1.1     3   54    64   163 28.163 -190065 68.071
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

R1-HQ#show logging | include Trap|Logging t
*Sep 26 00:49:44.543 AEST: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.4 on FastEthernet1/0 from FULL to DOWN, Neighbor Down: Dead timer expired|logging
Syslog logging: enabled (0 messages dropped, 2 messages rate-limited,
Console logging: level debugging, 123 messages logged, xml disabled,
Monitor logging: level debugging, 0 messages logged, xml disabled,
Buffer logging: level informational, 18 messages logged, xml disabled,
Count and timestamp logging messages: disabled
Persistent logging: disabled
Trap logging: level informational, 52 message lines logged
Logging to 10.30.100.10 (udp port 514, audit disabled,
R1-HQ#show running-config | section line vty
line vty 0 4
 access-class MGMT-ACCESS in
 login local
 transport input ssh
R1-HQ#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 9: R1-HQ – SSH v2, NTP synchronisation to R4-CORE, remote Syslog and VTY hardening

```

Part A | PC-IT1 (HQ VLAN 10) | DHCP lease, DNS, inter-site + Internet connectivity [192.168.56.104:5015] 2026-09-25 14:52:54
=====
PC-IT1:/#ip -4 addr show eth0 | grep inet
inet 10.10.10.24/24 scope global eth0
PC-IT1:/#cat /etc/resolv.conf
nameserver 10.0.0.3
PC-IT1:/#nslookup web.corp.local 2>/dev/null | grep -A1 Name
Name:    web.corp.local
Address: 10.30.60.10
PC-IT1:/#ping -c 2 -W 2 10.30.60.10
PING 10.30.60.10 (10.30.60.10): 56 data bytes
64 bytes from 10.30.60.10: seq=0 ttl=61 time=71.638 ms
64 bytes from 10.30.60.10: seq=1 ttl=61 time=71.278 ms

--- 10.30.60.10 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 71.278/71.458/71.638 ms
PC-IT1:/#ping -c 2 -W 2 10.20.40.22
PING 10.20.40.22 (10.20.40.22): 56 data bytes
64 bytes from 10.20.40.22: seq=0 ttl=61 time=129.971 ms
64 bytes from 10.20.40.22: seq=1 ttl=61 time=89.125 ms

--- 10.20.40.22 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 89.125/109.548/129.971 ms
PC-IT1:/#ping -c 2 -W 2 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: seq=0 ttl=58 time=78.196 ms
64 bytes from 8.8.8.8: seq=1 ttl=58 time=65.113 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 65.113/71.654/78.196 ms
PC-IT1:/#traceroute -n -w 2 -q 1 10.20.40.22
traceroute to 10.20.40.22 (10.20.40.22), 30 hops max, 46 byte packets
 1 10.10.10.1 46.052 ms
 2 10.255.0.1 56.190 ms
 3 10.255.0.6 69.202 ms
 4 10.20.40.22 64.870 ms
PC-IT1:/#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 10: PC-IT1 – DHCP lease, DNS resolution of web.corp.local, inter-site and Internet reachability, traceroute via the core

```

Part A | PC-GUEST1 (HQ VLAN 30) | GUEST-IN ACL - Internet allowed, internal networks blocked [192.168.56.104:5005] 2026-09-25 15:03:31
=====
PC-GUEST1:/#ip -4 addr show eth0 | grep inet
    inet 10.10.30.22/24 scope global eth0
PC-GUEST1:/#ping -c 2 -W 2 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: seq=0 ttl=58 time=83.547 ms
64 bytes from 8.8.8.8: seq=1 ttl=58 time=55.480 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 55.480/69.513/83.547 ms
PC-GUEST1:/#nslookup google.com 2>/dev/null | grep -A1 Name
Name:    google.com
Address: 142.250.183.46
PC-GUEST1:/#ping -c 2 -W 2 10.30.60.10
PING 10.30.60.10 (10.30.60.10): 56 data bytes

--- 10.30.60.10 ping statistics ---
2 packets transmitted, 0 packets received, 100% packet loss
PC-GUEST1:/#ping -c 2 -W 2 10.10.20.23
PING 10.10.20.23 (10.10.20.23): 56 data bytes

--- 10.10.20.23 ping statistics ---
2 packets transmitted, 0 packets received, 100% packet loss
PC-GUEST1:/#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation> |

```

Figure 11: PC-GUEST1 – Internet and DNS allowed; internal servers and Finance blocked by GUEST-IN

```

Part A | PC-SALES1 (Branch VLAN 40) | Web server reachable, Finance VLAN blocked by ACL [192.168.56.104:5007] 2026-09-25 15:05:32
=====
PC-SALES1:/#ip -4 addr show eth0 | grep inet
    inet 10.20.40.22/24 scope global eth0
PC-SALES1:/#nslookup web.corp.local 2>/dev/null | grep -A1 Name
Name:    web.corp.local
Address: 10.30.60.10
PC-SALES1:/#wget -qO- http://10.30.60.10 | grep -i title
<title>Welcome to nginx!</title>
PC-SALES1:/#ping -c 2 -W 2 10.10.10.24
PING 10.10.10.24 (10.10.10.24): 56 data bytes
64 bytes from 10.10.10.24: seq=0 ttl=61 time=60.677 ms
64 bytes from 10.10.10.24: seq=1 ttl=61 time=62.987 ms

--- 10.10.10.24 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 60.677/61.832/62.987 ms
PC-SALES1:/#ping -c 2 -W 2 10.10.20.23
PING 10.10.20.23 (10.10.20.23): 56 data bytes

--- 10.10.20.23 ping statistics ---
2 packets transmitted, 0 packets received, 100% packet loss
PC-SALES1:/#traceroute -I -n -w 2 -q 1 10.30.60.10
traceroute to 10.30.60.10 (10.30.60.10), 30 hops max, 46 byte packets
 1 10.20.40.1 41.194 ms
 2 10.255.0.5 61.104 ms
 3 10.255.0.10 71.679 ms
 4 10.30.60.10 51.758 ms
PC-SALES1:/#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation> |

```

Figure 12: PC-SALES1 – web server (HTTP) reachable, Finance VLAN blocked by FINANCE-PROTECT

```

Part A | R1-HQ | ACL hit count
=====
Part A | R1-HQ | ACL hit counters after connectivity tests [localhost:5001] 2026-09-25 15:06:27
=====

R1-HQ#show ip access-lists
Standard IP access list MGMT-ACCESS
 10 permit 10.30.100.0, wildcard bits 0.0.0.255
 20 permit 10.10.10.0, wildcard bits 0.0.0.255
 30 deny any log
Extended IP access list FINANCE-PROTECT
 10 permit ip 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255
 20 permit ip 10.30.0.0 0.0.255.255 10.10.20.0 0.0.0.255
 30 permit ip 10.0.0.0 0.0.0.255 10.10.20.0 0.0.0.255
 40 deny ip 10.20.0.0 0.0.255.255 10.10.20.0 0.0.0.255 log (6 matches)
 50 deny ip 10.10.30.0 0.0.0.255 10.10.20.0 0.0.0.255 log
 60 permit ip any any
Extended IP access list GUEST-IN
 10 permit udp any any eq bootps (24 matches)
 20 permit udp 10.10.30.0 0.0.0.255 host 10.0.3 eq domain (8 matches)
 30 permit icmp 10.10.30.0 0.0.0.255 host 10.10.30.1
 40 deny ip 10.10.30.0 0.0.0.255 10.0.0.0 0.255.255.255 log (8 matches)
 50 permit ip 10.10.30.0 0.0.0.255 any (4 matches)
R1-HQ#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 13: R1-HQ – ACL hit counters after the tests prove the policy is enforced

```

LIVE - ANSIBLE-SRV (Ansible)
messages: 978 from 10 devices: 10.0.0.1 10.0.0.2 10.0.0.3 10.0.0.4 10.10.99.11 10.10.99.12 10.20.99.11 10.20.99.12 10.30.99.11 10.30.99.12

2026-09-25 19:06:51 10.20.99.12 NOTICE 81: .Sep 25 19:04:20.615 AEST: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: netadmin] [Source: 10.30.100.10] [localport: 22] at 19:04:20 AEST Fr
2026-09-25 19:08:26 10.0.0.2 NOTICE 33: Sep 25 19:08:26.187 AEST: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: netadmin] [Source: 10.30.100.10] [localport: 22] at 19:08:26 AEST Fri
2026-09-25 19:10:55 10.0.0.2 NOTICE 34: Sep 25 19:10:53.675 AEST: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: netadmin] [Source: 10.30.100.10] [localport: 22] at 19:10:53 AEST Fri
2026-09-25 19:11:23 10.20.99.12 NOTICE 82: .Sep 25 19:08:43.879 AEST: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: netadmin] [Source: 10.30.100.10] [localport: 22] at 19:08:43 AEST Fr
2026-09-25 19:11:23 10.20.99.11 NOTICE 82: .Sep 25 19:08:43.481 AEST: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: netadmin] [Source: 10.30.100.10] [localport: 22] at 19:08:43 AEST Fr
2026-09-25 19:11:28 10.20.99.11 NOTICE 83: .Sep 25 19:08:40.357 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:vlan database
2026-09-25 19:11:28 10.20.99.12 NOTICE 83: .Sep 25 19:08:40.891 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:vlan database
2026-09-25 19:11:30 10.20.99.11 NOTICE 84: .Sep 25 19:08:50.557 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:vlan 45 name MARKETING
2026-09-25 19:11:31 10.20.99.12 NOTICE 84: .Sep 25 19:08:51.227 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:vlan 45 name MARKETING
2026-09-25 19:11:32 10.20.99.11 NOTICE 85: .Sep 25 19:07:03.933 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:apply
2026-09-25 19:11:33 10.20.99.12 NOTICE 85: .Sep 25 19:08:53.263 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:apply
2026-09-25 19:13:49 10.20.99.11 NOTICE 86: .Sep 25 19:08:24.529 AEST: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: netadmin] [Source: 10.30.100.10] [localport: 22] at 19:08:24 AEST Fr
2026-09-25 19:13:58 10.20.99.11 NOTICE 87: .Sep 25 19:09:29.249 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:spanning-tree vlan 45 priority 4096
2026-09-25 19:13:58 10.20.99.11 NOTICE 88: .Sep 25 19:09:29.561 AEST: %SYS-5-CONFIG_I: Configured from console by netadmin on vty0 (10.30.100.10)
2026-09-25 19:14:02 10.20.99.12 NOTICE 86: .Sep 25 19:09:33.669 AEST: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: netadmin] [Source: 10.30.100.10] [localport: 22] at 19:09:33 AEST Fr
2026-09-25 19:14:25 10.0.0.2 NOTICE 35: Sep 25 19:14:24.903 AEST: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: netadmin] [Source: 10.30.100.10] [localport: 22] at 19:14:24 AEST Fri
2026-09-25 19:14:31 10.0.0.2 NOTICE 36: Sep 25 19:14:30.839 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:interface FastEthernet0/0.45
2026-09-25 19:14:31 10.0.0.2 NOTICE 37: Sep 25 19:14:31.251 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:description VLAN45-MARKETING
2026-09-25 19:14:31 10.0.0.2 NOTICE 38: Sep 25 19:14:31.611 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:encapsulation dot1Q 45
2026-09-25 19:14:33 10.0.0.2 NOTICE 39: Sep 25 19:14:32.123 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:ip address 10.20.45.1 255.255.255.0
2026-09-25 19:14:33 10.0.0.2 NOTICE 40: Sep 25 19:14:32.343 AEST: %SYS-5-CONFIG_I: Configured from console by netadmin on vty0 (10.30.100.10)
2026-09-25 19:14:43 10.0.0.2 NOTICE 41: Sep 25 19:14:42.275 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:router ospf 1
2026-09-25 19:14:43 10.0.0.2 NOTICE 42: Sep 25 19:14:42.723 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:network 10.20.45.0 0.0.0.255 area 2
2026-09-25 19:14:43 10.0.0.2 NOTICE 43: Sep 25 19:14:42.847 AEST: %SYS-5-CONFIG_I: Configured from console by netadmin on vty0 (10.30.100.10)
2026-09-25 19:14:59 10.0.0.2 NOTICE 44: Sep 25 19:14:58.923 AEST: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: netadmin] [Source: 10.30.100.10] [localport: 22] at 19:14:58 AEST Fri
2026-09-25 19:15:10 10.0.0.2 NOTICE 45: Sep 25 19:15:08.519 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:ip dhcp excluded-address 10.20.45.1 10.20.45.20
2026-09-25 19:15:10 10.0.0.2 NOTICE 46: Sep 25 19:15:08.863 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:ip dhcp pool BR-MARKETING
2026-09-25 19:15:10 10.0.0.2 NOTICE 47: Sep 25 19:15:09.251 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:network 10.20.45.0 255.255.255.0
2026-09-25 19:15:10 10.0.0.2 NOTICE 48: Sep 25 19:15:09.555 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:default-router 10.20.45.1
2026-09-25 19:15:12 10.0.0.2 NOTICE 49: Sep 25 19:15:09.795 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:dns-server 10.0.0.3
2026-09-25 19:15:12 10.0.0.2 NOTICE 50: Sep 25 19:15:10.127 AEST: %PARSER-5-CFGLOG_LOGGEDCMD: User:netadmin logged command:domain-name corp.local
2026-09-25 19:15:12 10.0.0.2 NOTICE 51: Sep 25 19:15:10.391 AEST: %SYS-5-CONFIG_I: Configured from console by netadmin on vty0 (10.30.100.10)

=== STEP COMPLETE ===
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 14: ANSIBLE-SRV – central Syslog collector receiving messages from all network devices

3.5 Design justification

- **Hierarchical model:** separating core, distribution and access functions keeps failure domains small, makes behaviour predictable and lets the design scale. A new site is simply another distribution block attached to the core.
- **Multi-area OSPF:** an open standard that converges quickly. ABR summarisation shrinks the backbone table to three /16 routes and keeps site flaps out of the other areas' SPF runs. MD5

authentication blocks rogue neighbours, and passive-interface default stops hellos on user VLANs.

- **Path redundancy:** the R1–R3 link (OSPF cost 100) is a backup HQ↔DC path. Dual trunks with a deterministic STP root give Layer 2 redundancy without loops.
- **Router-on-a-stick:** suits the emulated EtherSwitch platform and centralises inter-VLAN policy (ACLs) at a single gateway per site.
- **Local DHCP and central DNS:** each branch keeps working if the WAN fails, while name resolution stays consistent from one corp.local zone in the DC.
- **Management plane:** SSH v2 only, with an ACL restricting management to the NetOps and IT VLANs. NTP gives consistent timestamps, and Syslog plus archive logging give a central audit trail.
- **Segmentation and least privilege:** Guest users get Internet access only, Finance is isolated from branch and guest users, and the server VLAN admits only the services it needs.
- **Single Internet edge with PAT:** private addressing internally, with one controlled, auditable egress point.

4. Network Automation with Ansible (Part B)

4.1 Automation architecture

ANSIBLE-SRV (10.30.100.10, NETOPS VLAN 100) is the Linux control node. It reaches every router loopback and switch management SVI in-band over OSPF, using SSH (network_cli with paramiko) and privilege escalation through enable. The Ansible project is stored in Git and mounted on the server at /root/ansible. The server also runs a small Python Syslog collector (tools/syslog_server.py) that receives logs from every device.

Element	Implementation
Inventory	inventory/hosts.yml – groups by role (core_routers, site_routers, distribution_switches, access_switches) and by site (site_hq, site_branch, site_dc, site_core)
Source of truth	group_vars/all (standards: NTP, DNS, Syslog, banner, SNMPv3, mgmt ACL), group_vars/site_* (VLANs, ACLs, DNS records, NAT), host_vars/* (router-IDs, P2P links, trunks, access ports)
Secrets	group_vars/all/vault.yml encrypted with ansible-vault (AES-256); .vault_pass is git-ignored
Roles	backup, baseline, vlans, routing, services, acl, verify, compliance
Templates	Jinja2: interfaces.j2, ospf.j2, services.j2, report.md.j2, compliance.md.j2
Playbooks	00_facts_inventory ... 08_compliance, site.yml (full pipeline)
Outputs	backups/ (timestamped + latest), reports/ (inventory CSV, verification, compliance), logs/ansible.log

Table 6: Ansible project components

```

=====
LIVE CONSOLE | ANSIBLE-SRV - Ansible control node (10.30.100.10) [192.168.56.104:5001]
=====
>>> Part B | Ansible control node: version, collections, inventory graph

((venv) ) root@ANSIBLE-SRV:~/ansible# ansible --version | head -4; echo; ansible-galaxy collection list 2>/dev/null | grep -E 'cisco|ansible\.'; echo; ansible-invent
ory --graph routers; ansible-inventory --graph switches
ansible [core 2.17.14]
  config file = /root/ansible/ansible.cfg
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /root/ansible/venv/lib/python3.12/site-packages/ansible

ansible.netcommon 8.7.1
ansible.posix     2.2.2
ansible.utils     6.1.1
cisco.ios         11.6.0
ansible.netcommon 8.7.1
ansible.posix     2.2.2
ansible.utils     6.1.1
cisco.ios         11.6.0

@routers:
  |--@core_routers:
  | |--R4-CORE
  |--@site_routers:
  | |--R1-HQ
  | |--R2-BR
  | |--R3-DC
@switches:
  |--@distribution_switches:
  | |--SW1-HQ-DIST
  | |--SW3-BR-DIST
  | |--SW5-DC-DIST
  |--@access_switches:
  | |--SW2-HQ-ACC
  | |--SW4-BR-ACC
  | |--SW6-DC-ACC
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 15: Ansible project tree, version and inventory graph on the automation server

4.2 Playbooks and roles

Playbook	Role	Target	What it does
00_facts_inventory.yml	–	all	Gathers ios_facts; writes reports/device_inventory.csv

Playbook	Role	Target	What it does
01_backup.yml	backup	all	Timestamped running-config backups + backups/latest for Git diff
02_baseline.yml	baseline	all	Hardening, local user, NTP, Syslog, DNS, archive logging, MGMT ACL, VTY, banner, SNMPv3
03_switching.yml	vlangs	switches	Creates missing VLANs, trunks, access ports, STP root, VLAN 99 SVI
04_routing.yml	routing	routers	Loopbacks, sub-interfaces, P2P links, multi-area OSPF (Jinja2)
05_services.yml	services	routers	DHCP pools, DNS host records, NAT/PAT (Jinja2)
06_security_acl.yml	acl	routers	Data-driven extended ACLs + hit counters
07_verify.yml	verify	all	Asserts OSPF FULL, default route, VLANs, trunks, pings; writes verification report
08_compliance.yml	compliance	all	14 security rules checked by regex; writes compliance report
site.yml	all	all	Runs the whole pipeline in order

Table 7: Playbooks and roles

Source of truth – ansible/inventory/group_vars/site_hq.yml (excerpt)

```

---
site_name: HQ
ospf_area: 10
site_summary: {prefix: 10.10.0.0, mask: 255.255.0.0}
mgmt_gateway: 10.10.99.1

vlans:
  - {id: 10, name: IT,          network: 10.10.10.0, gateway: 10.10.10.1, dhcp: true}
  - {id: 20, name: FINANCE, network: 10.10.20.0, gateway: 10.10.20.1, dhcp: true, acl_out:
    FINANCE-PROTECT}
  - {id: 30, name: GUEST,     network: 10.10.30.0, gateway: 10.10.30.1, dhcp: true, acl_in:
    GUEST-IN}
  - {id: 99, name: MGMT,      network: 10.10.99.0, gateway: 10.10.99.1, dhcp: false}

```

Jinja2 template – ansible/roles/routing/templates/ospf.j2

```

{# Multi-area OSPF: area 0 backbone on P2P links + loopbacks; site LANs in the site area,
   summarised at the ABR with 'area X range'. MD5 authentication on area 0. #}
router ospf {{ ospf.process_id }}
router-id {{ router_id }}
log-adjacency-changes
auto-cost reference-bandwidth {{ ospf.reference_bandwidth }}
area 0 authentication message-digest
{% if ospf_area is defined %}
  area {{ ospf_area }} range {{ site_summary.prefix }} {{ site_summary.mask }}
{% endif %}
passive-interface default
{% for l in p2p_links %}
  no passive-interface {{ l.interface }}
{% endfor %}
network {{ loopback0 }} 0.0.0.0 area 0
{% for l in p2p_links %}
  network {{ (l.ip ~ '/' ~ l.mask) | ansible.utils.ipaddr('network') }} 0.0.0.3 area 0
{% endfor %}
{% for v in vlans | default([]) %}
  network {{ v.network }} 0.0.0.255 area {{ ospf_area }}
{% endfor %}
{% if ntp_master | default(false) %}
  default-information originate always
{% endif %}

```

Role task – ansible/roles/vlans/tasks/main.yml (idempotent VLAN creation)

```

---
# VLAN + trunk/access port management for EtherSwitch (NM-16ESW) switches.
# NM-16ESW stores VLANs in 'vlan database' mode, so we read the current VLAN table and
# only create what is missing (idempotent), then manage ports with ios_config.
# Lines IOS never displays ('no shutdown', default 'switchport mode access') are not pushed,
# so repeated runs report changed=0.
- name: Read current VLAN table
  cisco.ios.ios_command:
    commands: show vlan-switch brief
  register: vlan_table
  changed_when: false

- name: Work out which VLANs are missing
  ansible.builtin.set_fact:
    missing_vlans: "{{ vlans | rejectattr('id', 'in', present_ids) | list }}"
  vars:
    present_ids: "{{ vlan_table.stdout[0] | regex_findall('(?!m)^(\\d+)\\s') | map('int') |
list }}"

- name: "Check mode: VLANs that WOULD be created"
  ansible.builtin.debug:
    msg: "would create: {{ missing_vlans | map(attribute='id') | zip(missing_vlans |
map(attribute='name')) | map('join', ' ') | list }}"
  when: ansible_check_mode and missing_vlans | length > 0
  changed_when: true

- name: Create missing VLANs in the VLAN database
  ansible.netcommon.cli_command:

```

4.3 Automation outputs

```

=====
LIVE CONSOLE | ANSIBLE-SRV - Ansible control node (10.30.100.10) [192.168.56.104:5001]
=====
>>> Part B | Ansible Vault - device credentials encrypted with AES-256 (values never shown)

((venv) ) root@ANSIBLE-SRV:~/ansible# rm -rf /root/.ansible/collections; ls -l inventory/group_vars/all/; echo; head -6 inventory/group_vars/all/vault.yml; echo '...';
echo; echo '# decrypted variable NAMES only:; ansible-vault view inventory/group_vars/all/vault.yml | sed -E 's/./: /'; echo; grep -n vault_inventory/
group_vars/all/main.yml roles/baseline/tasks/main.yml
total 8
-rw-rw-rw- 1 root root 1392 Sep 25 14:38 main.yml
-rw-rw-rw- 1 root root 1067 Sep 25 15:37 vault.yml

$ANSIBLE_VAULT;1.1;AES256
336534376631306362233835353265303836626639666135622323037366130636332336338646330
646262323864626233643437393736653732337343630370a633665323266383063613366393433
66363838666536343737343565366335613163356530646362376333238666466343838646531
3366313838393231660a62656137653762363263635303439663736383735636164323838666661
39313830643639653266633362333639333432326461366439393065343566623561323062666430
...

# decrypted variable NAMES only:
---
vault_admin_password: *****
vault_enable_password: *****
vault_ospf_key: *****
vault_snmp_auth: *****
vault_snmp_priv: *****

inventory/group_vars/all/main.yml:6:ansible_password: "{{ vault_admin_password }}"
inventory/group_vars/all/main.yml:9:ansible_become_password: "{{ vault_enable_password }}"
roles/baseline/tasks/main.yml:26: configured_password: "{{ vault_admin_password }}"
roles/baseline/tasks/main.yml:111: - "snmp-server user {{ snmp.user }} {{ snmp.group }} v3 auth sha {{ vault_snmp_auth }} priv aes 128 {{ vault_snmp_priv }}"
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 16: Ansible Vault – credentials encrypted with AES-256; only variable names are ever displayed

```

LIVE - ANSIBLE-SRV (Ansible)
ok: [SW5-DC-DIST] =>
msg: SW5-DC-DIST | 2691 | IOS 12.4(23) | tftp://255.255.255.255/unknown
ok: [SW4-BR-ACC] =>
msg: SW4-BR-ACC | 2691 | IOS 12.4(23) | tftp://255.255.255.255/unknown
ok: [SW2-HQ-ACC] =>
msg: SW2-HQ-ACC | 2691 | IOS 12.4(23) | tftp://255.255.255.255/unknown
ok: [SW6-DC-ACC] =>
msg: SW6-DC-ACC | 2691 | IOS 12.4(23) | tftp://255.255.255.255/unknown

TASK [Write inventory CSV] *****
Friday 25 September 2026 16:23:48 +1000 (0:00:10.200) 0:02:45.800 *****
ok: [R4-CORE -> localhost]

PLAY RECAP *****
R1-HQ : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
R2-BR : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
R3-DC : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
R4-CORE : ok=3 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW1-HQ-DIST : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW2-HQ-ACC : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW3-BR-DIST : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW4-BR-ACC : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW5-DC-DIST : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW6-DC-ACC : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

TASKS RECAP *****
Friday 25 September 2026 16:24:04 +1000 (0:00:15.862) 0:03:01.663 *****
=====
Gather IOS facts ----- 155.46s
Write inventory CSV ----- 15.86s
Print device summary ----- 10.19s

PLAYBOOK RECAP *****
Playbook run took 0 days, 0 hours, 3 minutes, 1 seconds

hostname,mgmt_ip,role,model,ios_version,image,serial
R1-HQ,10.0.0.1,router,7206VXR,15.0(1)M,tftp://255.255.255.255/unknown,4279256517
R2-BR,10.0.0.2,router,7206VXR,15.0(1)M,tftp://255.255.255.255/unknown,4279256517
R3-DC,10.0.0.3,router,7206VXR,15.0(1)M,tftp://255.255.255.255/unknown,4279256517
R4-CORE,10.0.0.4,router,7206VXR,15.0(1)M,tftp://255.255.255.255/unknown,4279256517
SW1-HQ-DIST,10.10.99.11,switch,2691,12.4(23),tftp://255.255.255.255/unknown,XXXXXXXXXX
SW2-HQ-ACC,10.10.99.12,switch,2691,12.4(23),tftp://255.255.255.255/unknown,XXXXXXXXXX
SW3-BR-DIST,10.20.99.11,switch,2691,12.4(23),tftp://255.255.255.255/unknown,XXXXXXXXXX
SW4-BR-ACC,10.20.99.12,switch,2691,12.4(23),tftp://255.255.255.255/unknown,XXXXXXXXXX
SW5-DC-DIST,10.30.99.11,switch,2691,12.4(23),tftp://255.255.255.255/unknown,XXXXXXXXXX
SW6-DC-ACC,10.30.99.12,switch,2691,12.4(23),tftp://255.255.255.255/unknown,XXXXXXXXXX
((venv) ) root@ANSIBLE-SRV:~/ansible$

```

Figure 17: 00_facts_inventory.yml – ios_facts discovery of all 10 devices and the generated inventory CSV

```

LIVE - ANSIBLE-SRV (Ansible)
LIVE CONSOLE | ANSIBLE-SRV - Ansible control node (10.30.100.10) [192.168.56.104:5001]
>>> Part B | 01_backup.yml - back up running-config of every device

((venv) ) root@ANSIBLE-SRV:~/ansible$ ansible-playbook playbooks/01_backup.yml 2>&1 | tail -30; ls -l backups/latest/; echo __DEMO_DONE__
ok: [SW4-BR-ACC] =>
msg: SW4-BR-ACC running-config saved to SW4-BR-ACC_20260925-173202.cfg
ok: [SW6-DC-ACC] =>
msg: SW6-DC-ACC running-config saved to SW6-DC-ACC_20260925-173202.cfg

PLAY RECAP *****
R1-HQ : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
R2-BR : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
R3-DC : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
R4-CORE : ok=5 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW1-HQ-DIST : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW2-HQ-ACC : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW3-BR-DIST : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW4-BR-ACC : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW5-DC-DIST : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SW6-DC-ACC : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

TASKS RECAP *****
Friday 25 September 2026 17:32:38 +1000 (0:00:01.476) 0:00:36.626 *****
=====
backup : Back up running-config (R4-CORE) ----- 23.67s
backup : Maintain a 'latest' copy for Git diffing ----- 0.86s
backup : Ensure backup directory exists on the automation server ----- 1.77s
backup : Show backup location ----- 1.48s
backup : Create timestamp for this backup run ----- 0.70s

PLAYBOOK RECAP *****
Playbook run took 0 days, 0 hours, 0 minutes, 36 seconds

total 48
-rw-r----- 1 root root 4578 Sep 25 17:32 R1-HQ.cfg
-rw-r----- 1 root root 3189 Sep 25 17:32 R2-BR.cfg
-rw-r----- 1 root root 4149 Sep 25 17:32 R3-DC.cfg
-rw-r----- 1 root root 3303 Sep 25 17:32 R4-CORE.cfg
-rw-r----- 1 root root 2812 Sep 25 17:32 SW1-HQ-DIST.cfg
-rw-r----- 1 root root 2754 Sep 25 17:32 SW2-HQ-ACC.cfg
-rw-r----- 1 root root 2776 Sep 25 17:32 SW3-BR-DIST.cfg
-rw-r----- 1 root root 2696 Sep 25 17:32 SW4-BR-ACC.cfg
-rw-r----- 1 root root 2777 Sep 25 17:32 SW5-DC-DIST.cfg
-rw-r----- 1 root root 2701 Sep 25 17:32 SW6-DC-ACC.cfg
__DEMO_DONE__
((venv) ) root@ANSIBLE-SRV:~/ansible$

```

Figure 18: 01_backup.yml – timestamped running-configuration backups plus a 'latest' copy per device for Git diffing

```

LIVE - ANSIBLE-SRV (Ansible)
>>> Part B | 02_baseline.yml - security hardening, SSH, NTP, Syslog, DNS, SNMPv3, banner

((venv)) root@ANSIBLE-SRV:~/ansible# ansible-playbook playbooks/02_baseline.yml 2>&1 | tail -42; echo === STEP COMPLETE ===
skipping: [R2-BR]
skipping: [SW1-HQ-DIST]
skipping: [R3-DC]
skipping: [SW3-BR-DIST]
skipping: [SW5-DC-DIST]
skipping: [SW2-HQ-ACC]
skipping: [SW4-BR-ACC]
skipping: [SW6-DC-ACC]

PLAY RECAP *****
R1-HQ                : ok=12  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
R2-BR                : ok=12  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
R3-DC                : ok=11  changed=0  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
R4-CORE              : ok=12  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
SW1-HQ-DIST          : ok=12  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
SW2-HQ-ACC            : ok=12  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
SW3-BR-DIST          : ok=12  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
SW4-BR-ACC            : ok=12  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
SW5-DC-DIST          : ok=12  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
SW6-DC-ACC            : ok=12  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0

TASKS RECAP *****
Friday 25 September 2026  17:54:47 +1000 (0:00:00.700)    0:03:15.143 *****
=====
baseline : DNS client ----- 50.50s
baseline : Global security hardening and services ----- 23.14s
baseline : NTP client / master ----- 14.62s
baseline : Local administrator account (password set only when the account is created) -- 13.25s
baseline : Harden VTY lines (SSH only, ACL, timeout) ----- 13.13s
baseline : SNMPv3 (authPriv) read-only monitoring ----- 12.64s
baseline : Legal login banner ----- 12.36s
baseline : Management-plane ACL (who may SSH to network devices) ----- 11.34s
baseline : Syslog to the NetOps collector + local buffer ----- 11.15s
baseline : Configuration-change auditing (archive log -> syslog) ----- 11.00s
baseline : Console line ----- 10.55s
baseline : Check whether the SNMPv3 user already exists ----- 10.52s
baseline : Create SNMPv3 user (only if missing - users are not shown in running-config) --- 0.70s

PLAYBOOK RECAP *****
Playbook run took 0 days, 0 hours, 3 minutes, 15 seconds

=== STEP COMPLETE ===
((venv)) root@ANSIBLE-SRV:~/ansible#

```

Figure 19: 02_baseline.yml – hardening, SSH, NTP, Syslog, DNS, archive logging, SNMPv3 and banner enforced on all devices

```

LIVE - ANSIBLE-SRV (Ansible)
>>> Part B | 03_switching.yml - VLANs, 802.1Q trunks, access ports, STP root (idempotent)

((venv)) root@ANSIBLE-SRV:~/ansible# ansible-playbook playbooks/03_switching.yml 2>&1 | tail -40; echo === STEP COMPLETE ===
Friday 25 September 2026  18:16:25 +1000 (0:00:11.254)    0:01:29.234 *****
skipping: [SW2-HQ-ACC]
skipping: [SW4-BR-ACC]
skipping: [SW6-DC-ACC]
ok: [SW3-BR-DIST]
ok: [SW5-DC-DIST]
ok: [SW1-HQ-DIST]

TASK [vlans : Management SVI] *****
Friday 25 September 2026  18:16:32 +1000 (0:00:07.396)    0:01:36.631 *****
ok: [SW5-DC-DIST]
ok: [SW4-BR-ACC]
ok: [SW6-DC-ACC]
ok: [SW3-BR-DIST]
ok: [SW1-HQ-DIST]
ok: [SW2-HQ-ACC]

PLAY RECAP *****
SW1-HQ-DIST          : ok=5    changed=0  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
SW2-HQ-ACC            : ok=5    changed=0  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
SW3-BR-DIST          : ok=5    changed=0  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
SW4-BR-ACC            : ok=5    changed=0  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
SW5-DC-DIST          : ok=5    changed=0  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
SW6-DC-ACC            : ok=5    changed=0  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0

TASKS RECAP *****
Friday 25 September 2026  18:16:43 +1000 (0:00:10.845)    0:01:47.476 *****
=====
vlans : Trunk ports (802.1Q uplinks) ----- 30.95s
vlans : Read current VLAN table ----- 37.00s
vlans : Access ports (end hosts) ----- 11.26s
vlans : Management SVI ----- 10.85s
vlans : STP root bridge for all site VLANs on the distribution switch --- 7.40s
vlans : Work out which VLANs are missing ----- 1.01s
vlans : Create missing VLANs in the VLAN database ----- 0.91s

PLAYBOOK RECAP *****
Playbook run took 0 days, 0 hours, 1 minutes, 47 seconds

=== STEP COMPLETE ===
((venv)) root@ANSIBLE-SRV:~/ansible#

```

Figure 20: 03_switching.yml – VLAN database, 802.1Q trunks, access ports and STP root managed on the six switches

```

=====
LIVE CONSOLE | ANSIBLE-SRV - Ansible control node (10.30.100.10) [192.168.56.104:5001]
=====
>>> Part B | 04_routing.yml - sub-interfaces, P2P links, multi-area OSPF (Jinja2 templates)

((venv) ) root@ANSIBLE-SRV:~/ansible# ansible-playbook playbooks/04_routing.yml 2>&1 | tail -30; echo === STEP COMPLETE ===
TASK [routing : Render + apply interface configuration (sub-interfaces, P2P links)] ***
Friday 25 September 2026  18:21:25 +1000 (0:00:00.156)          0:00:00.156 *****
ok: [R2-BR]
ok: [R1-HQ]
ok: [R4-CORE]
ok: [R3-DC]

TASK [routing : Render + apply OSPF configuration] *****
Friday 25 September 2026  18:21:41 +1000 (0:00:15.824)          0:00:15.981 *****
ok: [R4-CORE]
ok: [R2-BR]
ok: [R1-HQ]
ok: [R3-DC]

PLAY RECAP *****
R1-HQ                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
R2-BR                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
R3-DC                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
R4-CORE              : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

TASKS RECAP *****
Friday 25 September 2026  18:21:48 +1000 (0:00:07.275)          0:00:23.257 *****
=====
routing : Render + apply interface configuration (sub-interfaces, P2P links) -- 15.82s
routing : Render + apply OSPF configuration ----- 7.28s

PLAYBOOK RECAP *****
Playbook run took 0 days, 0 hours, 0 minutes, 23 seconds

=== STEP COMPLETE ===
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 21: 04_routing.yml – Jinja2-rendered sub-interfaces, P2P links and multi-area OSPF match the running network (changed=0)

```

=====
LIVE CONSOLE | ANSIBLE-SRV - Ansible control node (10.30.100.10) [192.168.56.104:5001]
=====
>>> Part B | 05_services.yml - DHCP pools, DNS records, NAT/PAT (Jinja2)

((venv) ) root@ANSIBLE-SRV:~/ansible# ansible-playbook playbooks/05_services.yml 2>&1 | tail -30; echo === STEP COMPLETE ===
Friday 25 September 2026  18:22:01 +1000 (0:00:00.227)          0:00:00.227 *****
[WARNING]: To ensure idempotency and correct diff the input configuration lines
should be similar to how they appear if present in the running configuration on
device including the indentation
ok: [R3-DC]
ok: [R4-CORE]
changed: [R2-BR]
changed: [R1-HQ]

RUNNING HANDLER [services : save config] *****
Friday 25 September 2026  18:23:09 +1000 (0:01:08.149)          0:01:08.377 *****
ok: [R2-BR]
ok: [R1-HQ]

PLAY RECAP *****
R1-HQ                : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
R2-BR                : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
R3-DC                : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
R4-CORE              : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

TASKS RECAP *****
Friday 25 September 2026  18:23:16 +1000 (0:00:06.266)          0:01:14.643 *****
=====
services : Render + apply DHCP / DNS / NAT services ----- 68.15s
services : save config ----- 6.27s

PLAYBOOK RECAP *****
Playbook run took 0 days, 0 hours, 1 minutes, 14 seconds

=== STEP COMPLETE ===
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 22: 05_services.yml – DHCP pools, DNS host records and NAT/PAT rendered from site data

```

=====
LIVE CONSOLE | ANSIBLE-SRV - Ansible control node (10.30.100.10) [192.168.56.104:5001]
=====
>>> Part B | 06_security_acl.yml - data-driven extended ACLs (declarative) + hit counters

((venv) ) root@ANSIBLE-SRV:~/ansible# ansible-playbook playbooks/06_security_acl.yml -l R1-HQ,R3-DC 2>&1 | grep -v -E '^ + ' | tail -40; echo === STEP COMPLETE ===
- ' 10 deny tcp any any eq telnet log'
- ' 20 deny tcp any any eq www log'
- ' 30 deny tcp any any eq 22 log'
- ' 40 permit tcp any any eq 22 log'
ok: [R3-DC] =>
msg:
- Standard IP access list MGMT-ACCESS
- ' 10 permit 10.30.100.0, wildcard bits 0.0.0.255 (26 matches)'
- ' 20 permit 10.10.10.0, wildcard bits 0.0.0.255'
- ' 30 deny any log'
- Extended IP access list SERVERS-PROTECT
- ' 10 permit tcp 10.0.0.0 0.255.255.255 host 10.30.60.10 eq www'
- ' 20 permit tcp 10.0.0.0 0.255.255.255 host 10.30.60.10 eq 443'
- ' 30 permit icmp 10.0.0.0 0.255.255.255 10.30.60.0 0.0.0.255'
- ' 40 permit icmp any 10.30.60.0 0.0.0.255 echo-reply'
- ' 50 permit tcp 10.30.100.0 0.0.0.255 10.30.60.0 0.0.0.255 eq 22'
- ' 60 deny ip 10.10.30.0 0.0.0.255 any log'
- ' 70 permit ip 10.30.0.0 0.0.0.255 any'
- ' 80 deny ip any any log'
- Extended IP access list sl_def_acl
- ' 10 deny tcp any any eq telnet log'
- ' 20 deny tcp any any eq www log'
- ' 30 deny tcp any any eq 22 log'
- ' 40 permit tcp any any eq 22 log'

PLAY RECAP *****
R1-HQ      : ok=3  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
R3-DC      : ok=3  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

TASKS RECAP *****
Friday 25 September 2026  18:29:25 +1000 (0:00:00.176)      0:00:16.298 *****
acl : Apply extended ACLs from site data (replace block when different) -- 13.45s
acl : Read ACL hit counters ----- 2.63s
acl : Show ACL counters ----- 0.18s

PLAYBOOK RECAP *****
Playbook run took 0 days, 0 hours, 0 minutes, 16 seconds

=== STEP COMPLETE ===
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 23: 06_security_acl.yml – declarative ACLs (whole block replaced only if the device differs) and live hit counters

```

=====
LIVE - ANSIBLE-SRV (Ansible) X + v
=====
((venv) ) root@ANSIBLE-SRV:~/ansible# ansible-playbook playbooks/07_verify.yml 2>&1 | tail -45; echo === STEP COMPLETE ===
Friday 25 September 2026  18:30:46 +1000 (0:00:13.873)      0:01:11.194 *****
ok: [R1-HQ]
ok: [R4-CORE]
ok: [R2-BR]
ok: [SW4-BR-ACC]
ok: [SW2-HQ-ACC]
ok: [SW5-DC-DIST]
ok: [R3-DC]
ok: [SW1-HQ-DIST]
ok: [SW3-BR-DIST]
ok: [SW6-DC-ACC]

TASK [Verify : Write verification report] *****
Friday 25 September 2026  18:30:47 +1000 (0:00:01.562)      0:01:12.757 *****
changed: [R4-CORE -> localhost]

PLAY RECAP *****
R1-HQ      : ok=5  changed=0  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0
R2-BR      : ok=5  changed=0  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0
R3-DC      : ok=5  changed=0  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0
R4-CORE     : ok=6  changed=1  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0
SW1-HQ-DIST : ok=4  changed=0  unreachable=0  failed=0  skipped=4  rescued=0  ignored=0
SW2-HQ-ACC  : ok=4  changed=0  unreachable=0  failed=0  skipped=4  rescued=0  ignored=0
SW3-BR-DIST : ok=4  changed=0  unreachable=0  failed=0  skipped=4  rescued=0  ignored=0
SW4-BR-ACC  : ok=4  changed=0  unreachable=0  failed=0  skipped=4  rescued=0  ignored=0
SW5-DC-DIST : ok=4  changed=0  unreachable=0  failed=0  skipped=4  rescued=0  ignored=0
SW6-DC-ACC  : ok=4  changed=0  unreachable=0  failed=0  skipped=4  rescued=0  ignored=0

TASKS RECAP *****
Friday 25 September 2026  18:30:52 +1000 (0:00:05.013)      0:01:17.771 *****
verify : Reachability tests from each router (DC web server, NetOps server, Internet) -- 23.27s
verify : Collect operational state (routers) ----- 16.13s
verify : Collect operational state (switches) ----- 14.95s
verify : Switch reachability to its gateway and the NTP master ----- 13.87s
verify : Write verification report ----- 5.01s
verify : Record NTP sync state ----- 1.56s
verify : Assert all site VLANs exist and trunks are up ----- 1.35s
verify : Assert default route learned / originated ----- 1.08s
verify : Assert every OSPF adjacency is FULL ----- 0.47s

PLAYBOOK RECAP *****
Playbook run took 0 days, 0 hours, 1 minutes, 17 seconds

=== STEP COMPLETE ===
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 24: 07_verify.yml – automated assertions: OSPF FULL, default route, VLANs, trunks and reachability on every device

```

=====
LIVE CONSOLE | ANSIBLE-SRV - Ansible control node (10.30.100.10) [192.168.56.104:5001]
=====
>>> Part B | reports/verification_report.md generated by the verify role

((venv) ) root@ANSIBLE-SRV:~/ansible# cat reports/verification_report.md; echo === STEP COMPLETE ===
# ENT-HYBRID-NET - Automated Verification Report

Generated by Ansible on 'ANSIBLE-SRV' at 2026-09-25 18:30:48 AEST

| Device | Role | OSPF neighbours FULL | NTP synchronised | Result |
|-----|-----|-----|-----|-----|
| R1-HQ | router | 2/2 | False | PASS |
| R2-BR | router | 1/1 | False | PASS |
| R3-DC | router | 2/2 | False | PASS |
| R4-CORE | router | 3/3 | True | PASS |
| SW1-HQ-DIST | switch | n/a | False | PASS |
| SW2-HQ-ACC | switch | n/a | False | PASS |
| SW3-BR-DIST | switch | n/a | False | PASS |
| SW4-BR-ACC | switch | n/a | False | PASS |
| SW5-DC-DIST | switch | n/a | False | PASS |
| SW6-DC-ACC | switch | n/a | False | PASS |

## Router reachability tests (source Loopback0)

| Router | Destination | Packet loss |
|-----|-----|-----|
| R1-HQ | 10.30.60.10 | 0% |
| R1-HQ | 10.30.100.10 | 0% |
| R1-HQ | 10.0.0.4 | 0% |
| R1-HQ | 8.8.8.8 | 0% |
| R2-BR | 10.30.60.10 | 0% |
| R2-BR | 10.30.100.10 | 0% |
| R2-BR | 10.0.0.4 | 0% |
| R2-BR | 8.8.8.8 | 0% |
| R3-DC | 10.30.60.10 | 0% |
| R3-DC | 10.30.100.10 | 0% |
| R3-DC | 10.0.0.4 | 0% |
| R3-DC | 8.8.8.8 | 0% |
| R4-CORE | 10.30.60.10 | 0% |
| R4-CORE | 10.30.100.10 | 0% |
| R4-CORE | 10.0.0.4 | 0% |
| R4-CORE | 8.8.8.8 | 0% |
=== STEP COMPLETE ===
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 25: Verification report rendered by the verify role (reports/verification_report.md)

```

((venv) ) root@ANSIBLE-SRV:~/ansible# ansible-playbook playbooks/08_compliance.yml 2>&1 | grep -E 'msg:|failed=' | tail -24; echo; cat reports/compliance_report.md;
echo === STEP COMPLETE ===
msg: R4-CORE compliance 14/14 ALL PASS
msg: R2-BR compliance 14/14 ALL PASS
msg: R1-HQ compliance 14/14 ALL PASS
msg: R3-DC compliance 14/14 ALL PASS
msg: SW3-BR-DIST compliance 14/14 ALL PASS
msg: SW1-HQ-DIST compliance 14/14 ALL PASS
msg: SW5-DC-DIST compliance 14/14 ALL PASS
msg: SW4-BR-ACC compliance 14/14 ALL PASS
msg: SW2-HQ-ACC compliance 14/14 ALL PASS
msg: SW6-DC-ACC compliance 14/14 ALL PASS
R1-HQ      : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
R2-BR      : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
R3-DC      : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
R4-CORE     : ok=5  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SW1-HQ-DIST : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SW2-HQ-ACC  : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SW3-BR-DIST : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SW4-BR-ACC  : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SW5-DC-DIST : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SW6-DC-ACC  : ok=4  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

# ENT-HYBRID-NET - Security Compliance Report

Generated by Ansible at 2026-09-25 18:37:25 AEST

| Rule | R1-HQ | R2-BR | R3-DC | R4-CORE | SW1-HQ-DIST | SW2-HQ-ACC | SW3-BR-DIST | SW4-BR-ACC | SW5-DC-DIST | SW6-DC-ACC |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| SSH version 2 only | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Telnet disabled on VTY | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| VTY protected by MGMT-ACCESS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Local AAA user | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Enable secret (not password) | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Password encryption | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Login brute-force protection | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Legal banner | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| NTP configured | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Remote syslog | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Config-change archive logging | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| SNMPv3 priv group | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| No SNMPv1/v2c communities | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| HTTP server disabled | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| **Score** | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |

=== STEP COMPLETE ===
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 26: 08_compliance.yml – 14 security rules audited on 10 devices: 14/14 on every device

```

=====
LIVE CONSOLE | ANSIBLE-SRV ~ Ansible control node (10.30.100.10) [192.168.56.104:5001]
=====
>>> Part B | Idempotency: full configuration pipeline (02-06) second run - PLAY RECAP

((venv) ) root@ANSIBLE-SRV:~/ansible# grep -a -B2 -A11 'PLAY RECAP' logs/ansible.log | tail -13 | sed 's/^.*n=ansible | //' ; echo ; echo 'Playbook run took 6 minutes
24 seconds: 5 playbooks, 10 devices, changed=0 unreachable=0 failed=0'; echo === STEP COMPLETE ===

PLAY RECAP *****
R1-HQ      : ok=18  changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
R2-BR      : ok=17  changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
R3-DC      : ok=17  changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
R4-CORE     : ok=17  changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
SW1-HQ-DIST : ok=17  changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
SW2-HQ-ACC  : ok=17  changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
SW3-BR-DIST : ok=17  changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
SW4-BR-ACC  : ok=17  changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
SW5-DC-DIST : ok=17  changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
SW6-DC-ACC  : ok=17  changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
TASKS RECAP *****

Playbook run took 6 minutes 24 seconds: 5 playbooks, 10 devices, changed=0 unreachable=0 failed=0
=== STEP COMPLETE ===
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 27: Idempotency – the complete configuration pipeline (playbooks 02–06) re-run: changed=0, failed=0 on all 10 devices

Day-2 change delivered as code (CHG-0001: VLAN 45 MARKETING at the Branch)

A realistic change request was implemented purely by editing the source of truth: two lines added to `group_vars/site_branch.yml` and committed to Git (Figure C04). The same pipeline then dry-ran the change with `--check --diff`, applied it, and the result was verified on the devices. Because the data model links everything, the single VLAN entry produced the VLAN on both Branch switches, the STP root priority, the R2 router-on-a-stick sub-interface, the OSPF area 20 network statement and a new DHCP pool, with no hand-typed CLI.

```

=====
LIVE CONSOLE | ANSIBLE-SRV - Ansible control node (10.30.100.10) [192.168.56.104:5001]
=====
>>> Part B | Day-2 CHG-0001 dry run: --check --diff on site_branch (nothing is applied)

((venv) ) root@ANSIBLE-SRV:~/ansible# ansible-playbook playbooks/03_switching.yml playbooks/04_routing.yml playbooks/05_services.yml -l site_branch --check --diff 2>
&1 | grep -E 'PLAY \[[changed:|would create|'+[+]]failed=' | tail -46; echo === STEP COMPLETE ===
PLAY [Part B-3 | Switching - VLANs, 802.1Q trunks, access ports, STP root] *****
changed: [SW4-BR-ACC] =>
  msg: 'would create: [''45 MARKETING']'
changed: [SW3-BR-DIST] =>
  msg: 'would create: [''45 MARKETING']'
changed: [SW3-BR-DIST]
SW3-BR-DIST      : ok=6  changed=2  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0
SW4-BR-ACC       : ok=6  changed=1  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
PLAY [Part B-4 | Routing - loopbacks, inter-VLAN sub-interfaces, P2P links, multi-area OSPF] ***
changed: [R2-BR]
R2-BR            : ok=2  changed=2  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
SW3-BR-DIST      : ok=6  changed=2  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0
SW4-BR-ACC       : ok=6  changed=1  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
PLAY [Part B-5 | Network services - DHCP pools, DNS server records, NAT/PAT] ***
changed: [R2-BR]
R2-BR            : ok=3  changed=3  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
SW3-BR-DIST      : ok=6  changed=2  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0
SW4-BR-ACC       : ok=6  changed=1  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
=== STEP COMPLETE ===
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 28: CHG-0001 dry run (--check --diff): Ansible predicts the changes (VLAN 45 would be created) without touching any device

```

=====
LIVE CONSOLE | ANSIBLE-SRV - Ansible control node (10.30.100.10) [192.168.56.104:5001]
=====
>>> Part B | Day-2 CHG-0001 APPLY: VLAN 45 MARKETING rolled out to the Branch (SW3, SW4, R2)

((venv) ) root@ANSIBLE-SRV:~/ansible# ansible-playbook playbooks/03_switching.yml playbooks/04_routing.yml playbooks/05_services.yml -l site_branch 2>&1 | grep -E 'P
LAY \[[TASK \[[changed:|failed=' | tail -46; echo === STEP COMPLETE ===
PLAY [Part B-3 | Switching - VLANs, 802.1Q trunks, access ports, STP root] *****
TASK [vlans : Read current VLAN table] *****
TASK [vlans : Work out which VLANs are missing] *****
TASK [vlans : Check mode: VLANs that WOULD be created] *****
TASK [vlans : Create missing VLANs in the VLAN database] *****
changed: [SW3-BR-DIST] => (item=vlan database)
changed: [SW4-BR-ACC] => (item=vlan database)
changed: [SW3-BR-DIST] => (item=vlan 45 name MARKETING)
changed: [SW4-BR-ACC] => (item=vlan 45 name MARKETING)
changed: [SW3-BR-DIST] => (item=apply)
changed: [SW4-BR-ACC] => (item=apply)
changed: [SW3-BR-DIST] => (item=exit)
changed: [SW4-BR-ACC] => (item=exit)
TASK [vlans : Trunk ports (802.1Q uplinks)] *****
TASK [vlans : Access ports (end hosts)] *****
TASK [vlans : STP root bridge for all site VLANs on the distribution switch] ***
changed: [SW3-BR-DIST]
TASK [vlans : Management SVI] *****
SW3-BR-DIST      : ok=7  changed=2  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
SW4-BR-ACC       : ok=6  changed=1  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
PLAY [Part B-4 | Routing - loopbacks, inter-VLAN sub-interfaces, P2P links, multi-area OSPF] ***
TASK [routing : Render + apply interface configuration (sub-interfaces, P2P links)] ***
changed: [R2-BR]
TASK [routing : Render + apply OSPF configuration] *****
changed: [R2-BR]
R2-BR            : ok=3  changed=2  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SW3-BR-DIST      : ok=7  changed=2  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
SW4-BR-ACC       : ok=6  changed=1  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
PLAY [Part B-5 | Network services - DHCP pools, DNS server records, NAT/PAT] ***
TASK [services : Render + apply DHCP / DNS / NAT services] *****
changed: [R2-BR]
R2-BR            : ok=5  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SW3-BR-DIST      : ok=7  changed=2  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
SW4-BR-ACC       : ok=6  changed=1  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
=== STEP COMPLETE ===
((venv) ) root@ANSIBLE-SRV:~/ansible#

```

Figure 29: CHG-0001 applied to the Branch devices (SW3-BR-DIST, SW4-BR-ACC, R2-BR) – failed=0

```

Part B | Day-2 CHG-0001 verified on R2-BR - Fa0/0.45 gateway, DHCP pool BR-MARKETING, OSPF area 20 [localhost:5003] 2026-09-25 19:15:32
=====
R2-BR#show ip interface brief | include 0/0. Loopback
FastEthernet0/0      unassigned    YES NVRAM  up            up
FastEthernet0/0.40   10.20.40.1    YES NVRAM  up            up
FastEthernet0/0.45   10.20.45.1    YES manual up            up
FastEthernet0/0.50   10.20.50.1    YES NVRAM  up            up
FastEthernet0/0.99   10.20.99.1    YES NVRAM  up            up
Loopback0            10.0.0.2      YES NVRAM  up            up
R2-BR#show running-config interface FastEthernet0/0.45
Building configuration...

Current configuration : 127 bytes
!
interface FastEthernet0/0.45
description VLAN45-MARKETING
encapsulation dot1Q 45
ip address 10.20.45.1 255.255.255.0
end

R2-BR#show ip dhcp pool BR-MARKETING

Pool BR-MARKETING :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 0
Pending event                     : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
10.20.45.1         10.20.45.1 - 10.20.45.254             0
R2-BR#show ip ospf interface brief | include 0.45
Fa0/0.45          1    20    10.20.45.1/24    10    DR    0/0
R2-BR#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 30: CHG-0001 verified on R2-BR: Fa0/0.45 gateway 10.20.45.1, DHCP pool BR-MARKETING, OSPF area 20

```

Part B | Day-2 CHG-0001 verified on SW4-BR-ACC - VLAN 45 MARKETING in VLAN database, allowed on trunks [localhost:5008] 2026-09-25 19:16:03
=====
SW4-BR-ACC#show vlan-switch brief

VLAN Name                Status    Ports
-----
1    default                active    Fa1/0, Fa1/3, Fa1/4, Fa1/7, Fa1/8, Fa1/9, Fa1/10, Fa1/11, Fa1/12, Fa1/13, Fa1/14, Fa1/15
40   SALES                    active    Fa1/5
45   MARKETING                active    Fa1/6
50   OPERATIONS               active
99   MGMT                     active
1002 fddi-default             active
1003 token-ring-default      active
1004 fddinet-default         active
1005 trnet-default            active
SW4-BR-ACC#show interfaces trunk | include Fa1
Fa1/1    on          802.lq      trunking    1
Fa1/2    on          802.lq      trunking    1
Fa1/1    1-1005
Fa1/2    1-1005
Fa1/1    1,40,45,50,99
Fa1/2    1,40,45,50,99
Fa1/1    1,40,45,50,99
Fa1/2    none
SW4-BR-ACC#
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 31: CHG-0001 verified on SW4-BR-ACC: VLAN 45 MARKETING present in the VLAN database and carried on the trunks

Generated reports. The verification and compliance roles render Markdown reports into `ansible/reports/`, and those reports are committed to Git as evidence:

```
# ENT-HYBRID-NET - Automated Verification Report

Generated by Ansible on `ANSIBLE-SRV` at 2026-09-25 18:30:48 AEST

| Device | Role | OSPF neighbours FULL | NTP synchronised | Result |
|---|---|---|---|
| R1-HQ | router | 2/2 | False | PASS |
| R2-BR | router | 1/1 | False | PASS |
| R3-DC | router | 2/2 | False | PASS |
| R4-CORE | router | 3/3 | True | PASS |
| SW1-HQ-DIST | switch | n/a | False | PASS |
| SW2-HQ-ACC | switch | n/a | False | PASS |
| SW3-BR-DIST | switch | n/a | False | PASS |
| SW4-BR-ACC | switch | n/a | False | PASS |
| SW5-DC-DIST | switch | n/a | False | PASS |
| SW6-DC-ACC | switch | n/a | False | PASS |

## Router reachability tests (source Loopback0)

| Router | Destination | Packet loss |
|---|---|
| R1-HQ | 10.30.60.10 | 0% |
| R1-HQ | 10.30.100.10 | 0% |
| R1-HQ | 10.0.0.4 | 0% |
| R1-HQ | 8.8.8.8 | 0% |
| R2-BR | 10.30.60.10 | 0% |
| R2-BR | 10.30.100.10 | 0% |
| R2-BR | 10.0.0.4 | 0% |
| R2-BR | 8.8.8.8 | 0% |
| R3-DC | 10.30.60.10 | 0% |
| R3-DC | 10.30.100.10 | 0% |
| R3-DC | 10.0.0.4 | 0% |
| R3-DC | 8.8.8.8 | 0% |
```

```
| R4-CORE | 10.30.60.10 | 0% |
| R4-CORE | 10.30.100.10 | 0% |
| R4-CORE | 10.0.0.4 | 0% |
| R4-CORE | 8.8.8.8 | 0% |
```

ansible/reports/compliance_report.md

```
# ENT-HYBRID-NET - Security Compliance Report
```

```
Generated by Ansible at 2026-09-25 18:37:25 AEST
```

```
| Rule | R1-HQ | R2-BR | R3-DC | R4-CORE | SW1-HQ-DIST | SW2-HQ-ACC | SW3-BR-DIST | SW4-BR-ACC |
| SW5-DC-DIST | SW6-DC-ACC | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| SSH version 2 only | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Telnet disabled on VTY | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| VTY protected by MGMT-ACCESS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| PASS |
| Local AAA user | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Enable secret (not password) | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| PASS |
| Password encryption | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Login brute-force protection | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| PASS |
| Legal banner | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| NTP configured | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Remote syslog | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| Config-change archive logging | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| PASS |
| SNMPv3 priv group | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| No SNMPv1/v2c communities | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| PASS |
| HTTP server disabled | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS | PASS |
| **Score** | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |
```

4.4 Infrastructure as Code for the network

Infrastructure as Code (IaC) means managing infrastructure through machine-readable definition files rather than interactive configuration, so the same engineering practices used for software apply to it: version control, peer review, testing and repeatable deployment [5], [6]. In this project the network's desired state is YAML data, templates turn that data into device configuration, and Ansible makes the devices match.

- **Declarative and idempotent:** running the full configuration pipeline a second time changes nothing (Figure B13 – changed=0 on all ten devices). Ansible compares the desired lines with the running-configuration and pushes only the differences. That makes runs safe to repeat and lets the same playbook detect and correct configuration drift.
- **Single source of truth:** a VLAN, ACL or DNS record is defined once. The same data feeds the switch VLAN database, the router sub-interface, the DHCP pool and the OSPF network statement, so these can no longer drift apart.
- **Version control and review:** every change is a Git commit (Figure C03). The Day-2 VLAN rollout was a four-line YAML change that could be reviewed, previewed with `--check --diff`, applied, verified and rolled back with `git revert`.

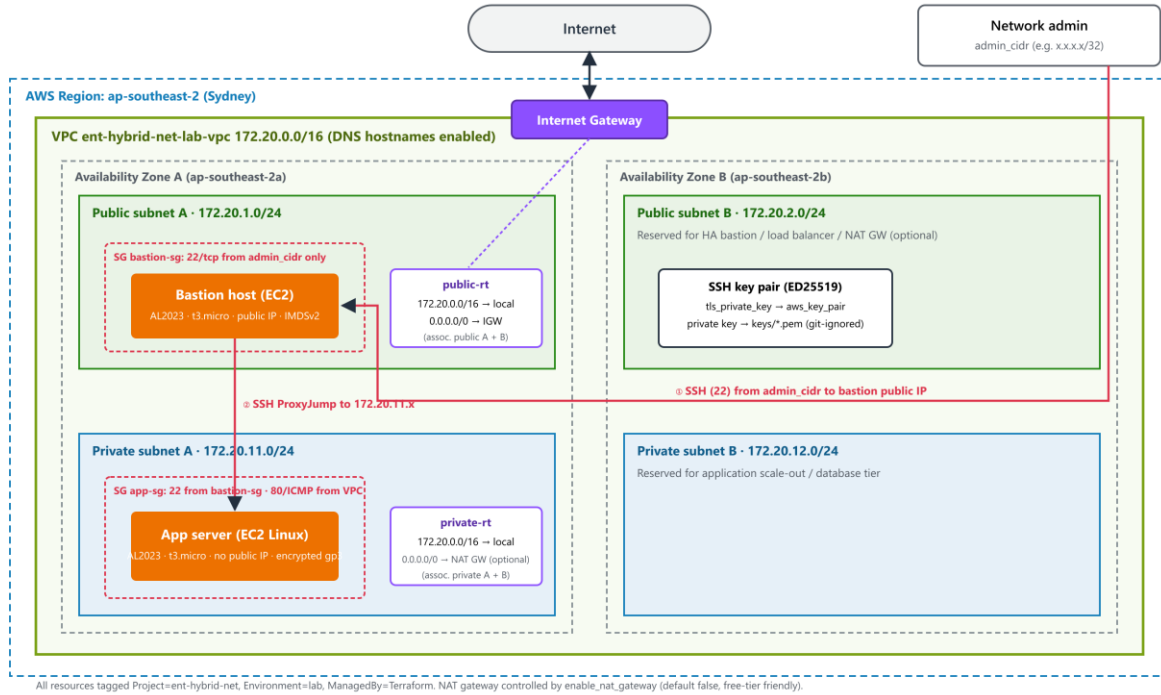
- **Security:** secrets are encrypted with Ansible Vault and never committed in plain text; the compliance role continuously audits the security baseline.
- **Speed and consistency:** the full pipeline configures and verifies all ten devices in minutes, whereas repeating the same work by hand at ten consoles is slow and error-prone.

5. AWS Infrastructure as Code with Terraform (Part C)

5.1 Architecture

ENT-HYBRID-NET – AWS Architecture (Terraform, ap-southeast-2)

VPC 172.20.0.0/16 · 2 AZs · 2 public + 2 private subnets · IGW · route tables · security groups · bastion + private EC2 · SSH key pair



ManagedBy=Terraform. Nothing sensitive is committed: terraform.tfvars, the state file, the plan file and the generated private key are all excluded by .gitignore, and a terraform.tfvars.example template is provided instead.

terraform/network.tf (excerpt)

```
data "aws_availability_zones" "available" {
  state = "available"
}

locals {
  azs = slice(data.aws_availability_zones.available.names, 0, 2)
  name = "${var.project}-${var.environment}"
}

# ----- VPC -----
resource "aws_vpc" "main" {
  cidr_block      = var.vpc_cidr
  enable_dns_support = true
  enable_dns_hostnames = true

  tags = { Name = "${local.name}-vpc" }
}

# ----- Internet Gateway -----
resource "aws_internet_gateway" "igw" {
  vpc_id = aws_vpc.main.id
  tags   = { Name = "${local.name}-igw" }
}

# ----- Subnets (2 public + 2 private across 2 AZs) -----
resource "aws_subnet" "public" {
  count          = 2
  vpc_id         = aws_vpc.main.id
  cidr_block     = var.public_subnet_cidrs[count.index]
  availability_zone = local.azs[count.index]
  map_public_ip_on_launch = true

  tags = { Name = "${local.name}-public-${local.azs[count.index]}", Tier = "public" }
}

resource "aws_subnet" "private" {
  count          = 2
  vpc_id         = aws_vpc.main.id
  cidr_block     = var.private_subnet_cidrs[count.index]
  availability_zone = local.azs[count.index]

  tags = { Name = "${local.name}-private-${local.azs[count.index]}", Tier = "private" }
}
```

terraform/security.tf (excerpt)

```
# ----- Bastion security group: SSH only from the admin's public IP -----
-
resource "aws_security_group" "bastion" {
  name          = "${local.name}-bastion-sg"
  description    = "SSH to bastion from admin CIDR only"
  vpc_id         = aws_vpc.main.id
  tags          = { Name = "${local.name}-bastion-sg" }
}

resource "aws_vpc_security_group_ingress_rule" "bastion_ssh" {
  security_group_id = aws_security_group.bastion.id
  description        = "SSH from admin"
}
```

```
cidr_ipv4      = var.admin_cidr
ip_protocol    = "tcp"
from_port      = 22
to_port        = 22
}

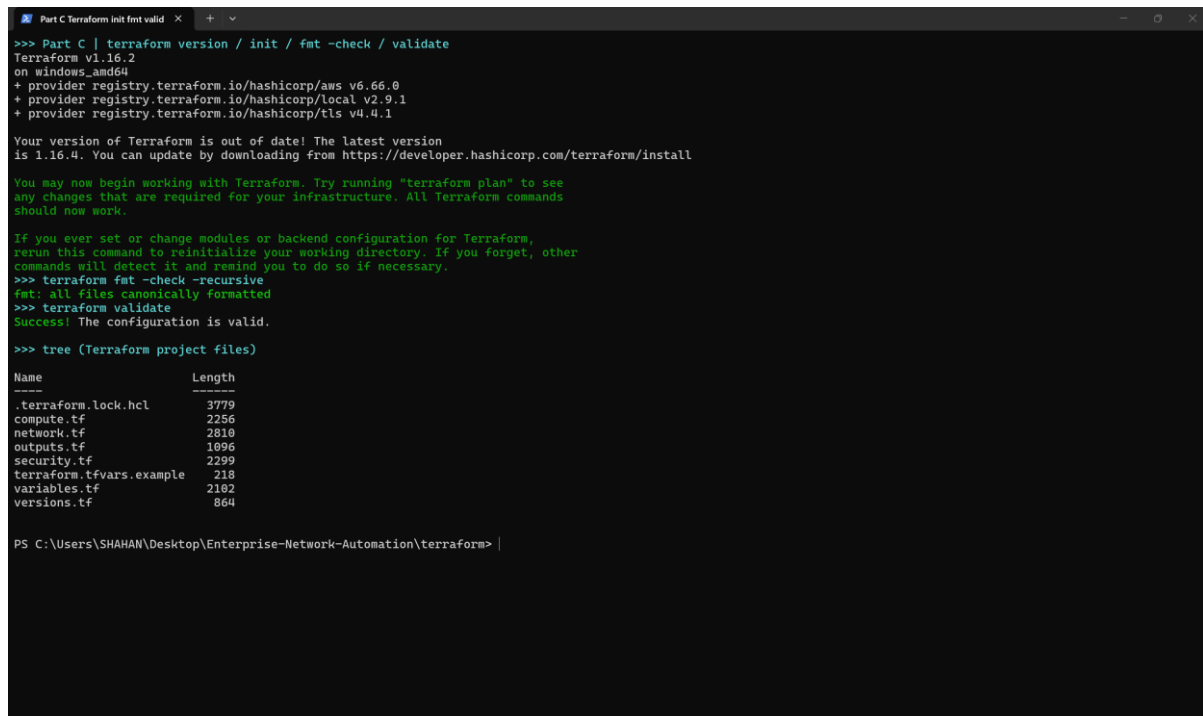
resource "aws_vpc_security_group_egress_rule" "bastion_all" {
  security_group_id = aws_security_group.bastion.id
  description       = "Allow all outbound"
```

5.3 Terraform workflow and AWS deployment

Terraform is declarative. The engineer describes the desired end state, and Terraform works out the create, update and delete actions needed by comparing the configuration with its state file and the real AWS resources [2]. The workflow used is:

1. `terraform init` – downloads the aws, tls and local providers and pins them in `.terraform.lock.hcl` (committed so every engineer uses identical provider versions).
2. `terraform fmt` and `terraform validate` – enforce canonical formatting and check syntax and references.
3. `terraform plan -out tfplan` – authenticates with the terraform-user IAM profile, reads the account (available AZs, latest AL2023 AMI) and prints an execution plan: 25 resources to add. The saved plan is what gets applied, so there are no surprises between review and deployment.
4. `terraform apply tfplan` – creates the resources in dependency order (VPC → subnets/IGW → route tables → SGs → key pair → instances) and prints the outputs, including a ready-made ProxyJump SSH command.
5. `terraform destroy` – removes everything, so lab costs stay at zero.

Deployment status. As agreed with the client, this engagement ran the workflow up to a reviewed, saved plan, and no AWS resources were created, to avoid cost. Running `terraform apply tfplan` followed by `terraform output` would deploy the environment exactly as planned.



```

Part C Terraform init fmt validate
>>> Part C | terraform version / init / fmt -check / validate
Terraform v1.16.2
on windows_amd64
+ provider registry.terraform.io/hashicorp/aws v6.66.0
+ provider registry.terraform.io/hashicorp/local v2.9.1
+ provider registry.terraform.io/hashicorp/tls v4.4.1

Your version of Terraform is out of date! The latest version
is 1.16.4. You can update by downloading from https://developer.hashicorp.com/terraform/install

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
>>> terraform fmt -check -recursive
fmt: all files canonically formatted
>>> terraform validate
Success! The configuration is valid.

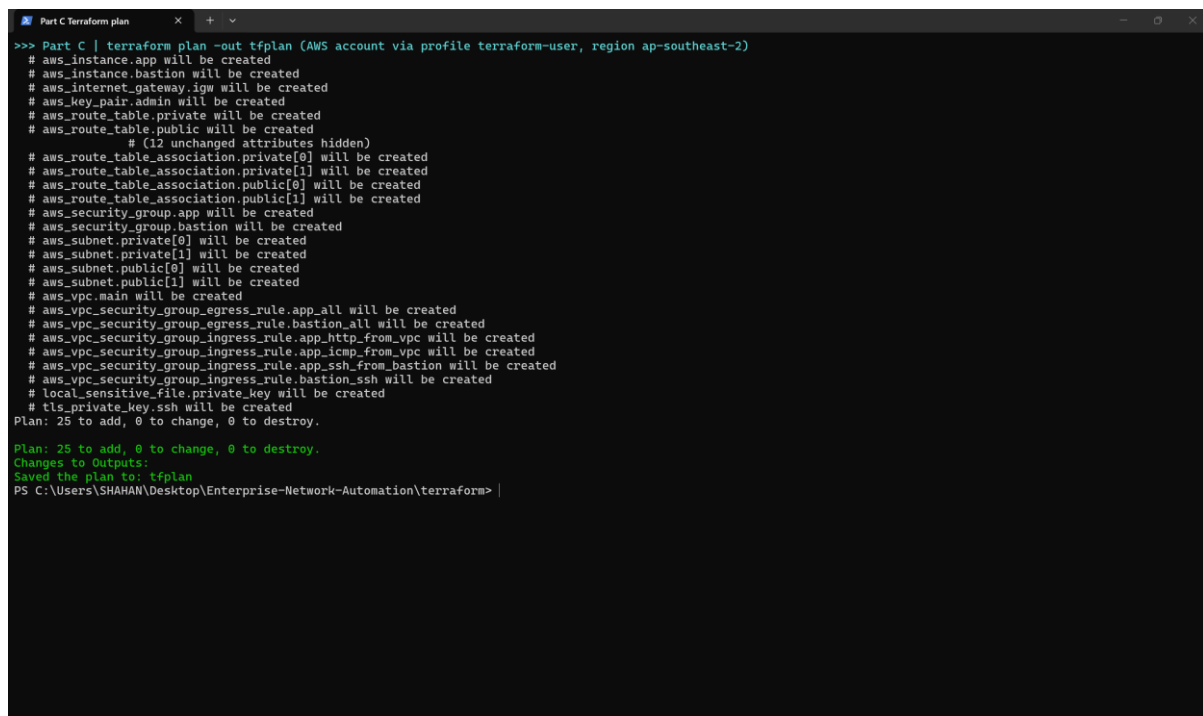
>>> tree (Terraform project files)

Name      Length
----      -
.terraform.lock.hcl  3779
compute.tf          2256
network.tf          2810
outputs.tf          1096
security.tf         2299
terraform.tfvars.example  210
variables.tf        2102
versions.tf         864

PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation\terraform>

```

Figure 34: terraform init / fmt / validate



```

Part C Terraform plan
>>> Part C | terraform plan -out tfplan (AWS account via profile terraform-user, region ap-southeast-2)
# aws_instance.app will be created
# aws_instance.bastion will be created
# aws_internet_gateway.igw will be created
# aws_key_pair.admin will be created
# aws_route_table.private will be created
# aws_route_table.public will be created
# (12 unchanged attributes hidden)
# aws_route_table_association.private[0] will be created
# aws_route_table_association.private[1] will be created
# aws_route_table_association.public[0] will be created
# aws_route_table_association.public[1] will be created
# aws_security_group.app will be created
# aws_security_group.bastion will be created
# aws_subnet.private[0] will be created
# aws_subnet.private[1] will be created
# aws_subnet.public[0] will be created
# aws_subnet.public[1] will be created
# aws_vpc.main will be created
# aws_vpc_security_group_egress_rule.app_all will be created
# aws_vpc_security_group_egress_rule.bastion_all will be created
# aws_vpc_security_group_ingress_rule.app_http_from_vpc will be created
# aws_vpc_security_group_ingress_rule.app_icmp_from_vpc will be created
# aws_vpc_security_group_ingress_rule.app_ssh_from_bastion will be created
# aws_vpc_security_group_ingress_rule.bastion_ssh will be created
# local_sensitive_file.private_key will be created
# tls_private_key.ssh will be created
Plan: 25 to add, 0 to change, 0 to destroy.

Plan: 25 to add, 0 to change, 0 to destroy.
Changes to Outputs:
Saved the plan to: tfplan
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation\terraform>

```

Figure 35: terraform plan – 25 resources to add, with outputs

Terraform plan – resource summary (evidence/terraform/terraform_plan_summary.txt)

```

# aws_instance.app will be created
# aws_instance.bastion will be created
# aws_internet_gateway.igw will be created
# aws_key_pair.admin will be created
# aws_route_table.private will be created
# aws_route_table.public will be created
# (12 unchanged attributes hidden)
# aws_route_table_association.private[0] will be created
# aws_route_table_association.private[1] will be created
# aws_route_table_association.public[0] will be created

```

```
# aws_route_table_association.public[1] will be created
# aws_security_group.app will be created
# aws_security_group.bastion will be created
# aws_subnet.private[0] will be created
# aws_subnet.private[1] will be created
# aws_subnet.public[0] will be created
# aws_subnet.public[1] will be created
# aws_vpc.main will be created
# aws_vpc_security_group_egress_rule.app_all will be created
# aws_vpc_security_group_egress_rule.bastion_all will be created
# aws_vpc_security_group_ingress_rule.app_http_from_vpc will be created
# aws_vpc_security_group_ingress_rule.app_icmp_from_vpc will be created
# aws_vpc_security_group_ingress_rule.app_ssh_from_bastion will be created
# aws_vpc_security_group_ingress_rule.bastion_ssh will be created
# local_sensitive_file.private_key will be created
# tls_private_key.ssh will be created
Plan: 25 to add, 0 to change, 0 to destroy.
```

5.4 Version control with Git

All code, device configurations, backups, reports and evidence are held in one Git repository, pushed to a private GitHub repository. Commits are small and descriptive, following the build order: Part A topology and configs, then the Terraform code layer by layer (providers → network → security → compute → outputs), then the Ansible project and its generated backups and reports. The Day-2 VLAN change is its own commit, so the network change can be traced and reverted.

```
>>> git log --graph --oneline --date=short (version control of the whole project)
* ff49f98 2026-09-25 Evidence: Part A/B/C screenshots, Terraform plan output, lab recovery and demo tooling <Shakil Ahammed>
* 6dea959 2026-09-25 Part B: automation outputs - config backups, inventory, verification and compliance reports, syslog <Shakil Ahammed>
* 4cd4e9f 2026-09-25 Part B: make every role idempotent and check-node safe on IOS 12.4/15.0 <Shakil Ahammed>
* f4afa596 2026-09-25 CHG-0001: add VLAN 45 MARKETING at the Branch Office <Shakil Ahammed>
* 1f6aa70 2026-09-25 Add placeholder B04 backup screenshot (run still in progress) <Shakil Ahammed>
* 56017ad 2026-09-25 Lab stability fixes, offline Ansible install and live demo tooling <Shakil Ahammed>
* 5c485c9 2026-09-25 Part B: switch control node to ansible-core 2.17 plus required collections <Shakil Ahammed>
* 5d8cb0f 2026-09-25 Part B: Ansible project (inventory, roles, playbooks, server tools) and report generator <Shakil Ahammed>
* b0a96f7 2026-09-25 Part A: evidence screenshots - OSPF, DHCP, ACLs, DNS, VLANs, STP, SSH/NTP/Syslog, host tests <Shakil Ahammed>
* cb05eee 2026-09-25 Tooling: evidence capture (window-only screenshots), clock sync, Ansible sync to automation server <Shakil Ahammed>
* 16eaa78 2026-09-25 Docs: IP addressing plan, design justification, AWS architecture diagram <Shakil Ahammed>
* 8667924 2026-09-25 Part C: Terraform outputs incl. ProxyJump SSH command <Shakil Ahammed>
* c86c4c4 2026-09-25 Part C: SSH key pair, bastion host and private EC2 Linux instance (IMDSv2, encrypted EBS) <Shakil Ahammed>
* 79bd415 2026-09-25 Part C: security groups - bastion SSH from admin CIDR, app SSH from bastion SG only <Shakil Ahammed>
* 43180a1 2026-09-25 Part C: VPC, 2 public + 2 private subnets, IGW, route tables, optional NAT GW <Shakil Ahammed>
* ee729e8 2026-09-25 Part C: Terraform providers, variables with validation, default tags <Shakil Ahammed>
* ceec71 2026-09-25 Part A: console bootstrap tooling (telnet client, parallel config push) <Shakil Ahammed>
* eb2dc73 2026-09-25 Part A: Day-0 config generator and rendered device configs (VLAN, OSPF, DHCP, SSH, ACL, NAT, DNS, NTP, Syslog) <Shakil Ahammed>
* b9fce75 2026-09-25 Part A: move end hosts to GNS3 VM, replace VPCS with Docker Linux hosts (DHCP fix) <Shakil Ahammed>
* b99c0bf 2026-09-25 Part A: GNS3 REST API client and hierarchical topology builder <Shakil Ahammed>
* 1624c4f 2026-09-25 Initial project structure, README and .gitignore <Shakil Ahammed>

commits: 21 branch: main tracked files: 163
PS C:\Users\SHAHAM\Desktop\Enterprise-Network-Automation>
```

Figure 36: Git commit history (git log --graph) demonstrating version control of the whole project

```

Part 8 Day-2 change CHG-000
>>> git log -1 --stat ; git show (Day-2 change as code)
commit f4fa596028cd1959b19a128e867f0b00be8e079b (HEAD -> main)
Author: Shaikil Ahammed <shaikil.shaham0@gmail.com>
Date:   Fri Sep 25 18:56:43 2026 +1000

    CHG-0001: add VLAN 45 MARKETING at the Branch Office

    Day-2 change delivered as code: one YAML entry in group_vars/site_branch.yml drives the VLAN on SW3/SW4, the R2 sub-interface Fa0/0.45, the DHCP pool BR-MARKETIN
    G and the OSPF area 20 network statement.

    Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>

ansible/inventory/group_vars/site_branch.yml | 2 ++
1 file changed, 2 insertions(+)
diff --git a/ansible/inventory/group_vars/site_branch.yml b/ansible/inventory/group_vars/site_branch.yml
index 6b05355..3882590 100644
--- a/ansible/inventory/group_vars/site_branch.yml
+++ b/ansible/inventory/group_vars/site_branch.yml
@@ -7,6 +7,8 @@ mgmt_gateway: 10.20.99.1
 vlans:
   - {id: 40, name: SALES,      network: 10.20.40.0, gateway: 10.20.40.1, dhcp: true}
   - {id: 50, name: OPERATIONS, network: 10.20.50.0, gateway: 10.20.50.1, dhcp: true}
+ # Day-2 change CHG-0001: new Marketing department at the Branch Office
+ - {id: 45, name: MARKETING,   network: 10.20.45.0, gateway: 10.20.45.1, dhcp: true}
   - {id: 99, name: MGMT,       network: 10.20.99.0, gateway: 10.20.99.1, dhcp: false}

acls: []
PS C:\Users\SHAHAN\Desktop\Enterprise-Network-Automation>

```

Figure 37: git show of the Day-2 change – a four-line YAML diff drives a multi-device network change

6. Testing

Testing was done at three levels: manual verification from device consoles, end-host functional tests, and automated assertions in the Ansible verify and compliance roles, which are safe to repeat after every change.

#	Test	Expected result	Result	Evidence
T1	OSPF adjacencies core ↔ sites	3 FULL neighbours on R4; 2 on R1/R3; 1 on R2	PASS	Fig. A02, A06, B10
T2	Route summarisation	R4 sees 10.10/16, 10.20/16, 10.30/16 as O IA	PASS	Fig. A02
T3	Default route / NAT	Hosts ping 8.8.8.8; NAT translations on R4	PASS	Fig. A10
T4	DHCP	PCs receive address, gateway, DNS 10.0.0.3, domain corp.local	PASS	Fig. A03, A06, A10
T5	DNS	web.corp.local → 10.30.60.10; google.com resolved via forwarder	PASS	Fig. A05, A10
T6	Inter-VLAN / inter-site	PC-IT1 reaches web server and Branch PC	PASS	Fig. A10
T7	GUEST-IN ACL	Guest reaches Internet; internal 10/8 blocked	PASS	Fig. A11, A13
T8	FINANCE-PROTECT ACL	Branch cannot reach Finance; HTTP to web allowed	PASS	Fig. A12, A13
T9	STP redundancy	One redundant uplink blocked per site	PASS	Fig. A08
T10	SSH / MGMT-ACCESS	SSH v2 only; only NETOPS/IT can log in; banner shown	PASS	Fig. A09, B18
T11	NTP	Clients synchronised to 10.0.0.4 (stratum 4)	PASS	Fig. A09
T12	Syslog	Collector receives messages from all devices	PASS	Fig. A14
T13	Ansible idempotency	Second run of playbooks 02-06: changed=0	PASS	Fig. B13
T14	Automated verification	All assertions pass on 10 devices	PASS	Fig. B10, B11
T15	Compliance audit	14/14 rules on every device	PASS	Fig. B12
T16	Day-2 change (VLAN 45)	Dry run predicts change; apply succeeds; VLAN/SVI/DHCP/OSPF present	PASS	Fig. B14-B17, C04
T17	Config backup	10 running-configs saved and versioned	PASS	Fig. B04
T18	Terraform validate/plan	Valid; 25 to add, 0 change, 0 destroy	PASS	Fig. C01, C02

Table 9: Test plan and results

7. Discussion

7.1 Challenges and how they were solved

- **VPCS DHCP incompatibility.** VPCS never completed DHCP with Cisco IOS: it drops the DHCPACK that IOS unicasts to the offered address. Debugging with `debug ip dhcp server packet` showed the ACK leaving R1. We replaced the VPCS hosts with lightweight Docker Linux containers (Alpine `udhcpd`), which also gave us realistic tools (`nslookup`, `wget`, `traceroute`) and a real `nginx` web server for the ACL tests.
- **Emulator CPU and timers.** A switch that was running while its `idle-PC` value was calculated never picked the value up and consumed a full CPU core. Emulation stalls caused OSPF dead-timer expiries. Restarting the node applied the `idle-PC`, and after that the adjacencies stayed stable. Device clocks also started about 10 hours off, which stopped NTP from converging; setting the clocks once let NTP take over.
- **EtherSwitch VLAN database.** NM-16ESW modules keep VLANs in the legacy `vlan` database mode, which the `cisco.ios` resource modules do not support. The `vlan` role therefore parses `show vlan-switch brief` and creates only the missing VLANs through `cli_command`, which keeps the role idempotent.
- **Legacy SSH algorithms.** IOS 12.4/15.0 offers only older key-exchange and cipher algorithms. Using the `paramiko` transport for `network_cli` kept compatibility without weakening the automation server's system-wide OpenSSH policy.
- **Internet egress in the lab.** Large downloads through the emulated path (R3 → R4 PAT → GNS3 NAT) timed out, and the IOS DNS forwarder could not resolve the PyPI CDN names. Ansible was therefore installed offline: wheels and Galaxy collection tarballs were downloaded on the host and copied into the container (`scripts/offline_install.py`). This is the same pattern used in air-gapped production networks.
- **Hypervisor instability.** The VirtualBox log showed the GNS3 VM running through Hyper-V (NEM: "VT-x is not available") because Windows virtualisation-based security is enabled. In this mode the VM froze roughly every 35 minutes. The team raised the GNS3 VM to 4 GB / 2 vCPU and wrote `scripts/recover_lab.py` (reset VM, re-open project, wait for OSPF, re-sync clocks), which cut recovery time to about five minutes. Disabling Hyper-V on the lab host is the permanent fix.
- **Idempotency on real IOS.** The first baseline runs reported changes every time. IOS hides default settings (`ip ssh authentication-retries 3`, logging trap informational, `exec-timeout 10 0`, DHCP lease 1, no shutdown), rewrites some commands (logging host X is shown as logging X) and pads ACL keywords (`deny ip`). The roles were corrected to push exactly what IOS stores, platform differences were moved into group variables (12.4 lacks 'notify syslog contenttype' and SNMPv3 AES), and ACLs became declarative blocks. After this, repeated runs report `changed=0`.
- **Check mode safety.** `cli_command` (used for the legacy VLAN database) has no check mode, and the write-memory handler failed during `--check`. The role now reports the VLANs it would create, and handlers are skipped in check mode, so a dry run never writes NVRAM.
- **Data-quality bug caught by automation.** A generated `host_vars` file for R2-BR had its single P2P link flattened into characters. Ansible stopped on the first invalid line ('interface F') before changing anything, showing why an automation pipeline must fail safely.
- **Secrets in evidence.** Bootstrap logs and backups contained credentials or reversible type-7 strings. The team masked secrets in logs, redacted type-7 strings in the committed backups, and scanned every commit for secret values before pushing.

7.2 Security considerations

Security is layered. The management plane is restricted to SSH v2 from two source VLANs, with login brute-force blocking, a legal banner, archive logging and SNMPv3 authPriv. The control plane uses OSPF MD5 authentication and passive interfaces. The data plane is segmented with VLANs and ACLs. In the cloud, the bastion is the only host with an inbound Internet rule, and only from one /32; the application server has no public IP and accepts SSH only from the bastion's security group. IMDSv2 and EBS encryption are enforced. Secrets never reach Git in plain text: Ansible Vault protects device credentials, and Terraform's private key, state and tfvars are git-ignored.

7.3 Limitations and future work

- Add a CI/CD pipeline (GitHub Actions) that runs ansible-lint, yamllint, terraform fmt/validate/tflint and checkov on every pull request, and deploys on merge.
- Use NetBox as a dynamic inventory and source of truth, and pyATS/Batfish for pre-change validation.
- Connect the Data Centre to the AWS VPC with a site-to-site IPsec VPN (both address spaces were chosen not to overlap) and extend OSPF/BGP into the cloud.
- Move Terraform state to an encrypted S3 backend with state locking so the team can collaborate safely.
- Replace the emulated router-on-a-stick with Layer 3 distribution switches and HSRP for gateway redundancy on production hardware.

8. Conclusion

The project met all three technical objectives. A hierarchical, redundant and segmented enterprise network was designed and built in GNS3, with every required service (VLANs, inter-VLAN routing, OSPF, DHCP, SSH, ACLs, NAT, DNS, NTP and Syslog) configured and verified end to end. The network is now managed as code: an Ansible project with a YAML source of truth, Jinja2 templates, eight roles and Vault-protected secrets backs up, configures, hardens, verifies and audits all ten devices idempotently, and a Day-2 change was delivered from a single reviewed Git commit. In the cloud, Terraform describes a secure two-AZ AWS VPC with a bastion-protected private EC2 instance, validated and planned against the live account and version-controlled in Git.

Together these show the shift the client asked for: from manual administration to automated, auditable and repeatable infrastructure delivery across on-premises and cloud environments.

References

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Appendix A – Submission Contents and Evidence Index

Deliverable	Location in repository
Technical report	report/ENT-HYBRID-NET_Technical_Report.docx
GNS3 project	gns3/ENT-HYBRID-NET.gns3project (portable export) + gns3/build_topology.py
Configuration files	configs/*.cfg (Day-0) and ansible/backups/latest/*.cfg (running configs)
Ansible project	ansible/
Terraform project	terraform/
Git repository link	[GitHub URL]
Evidence screenshots	screenshots/*.png (listed below)

Table 10: Submission contents

Figure	Caption	File
Figure 1	ENT-HYBRID-NET GNS3 topology – hierarchical core / distribution / access design	screenshots/A01_gns3_topology.png
Figure 2	R4-CORE – three FULL OSPF adjacencies, /16 inter-area summaries per site and the default route to the Internet	screenshots/A02_R4_ospf_neighbors_routes.png
Figure 3	R1-HQ – router-on-a-stick sub-interfaces and DHCP bindings for HQ clients	screenshots/A03_R1_interfaces_dhcp.png
Figure 4	R1-HQ – GUEST-IN, FINANCE-PROTECT and MGMT-ACCESS ACLs; OSPF interfaces in areas 0 and 10	screenshots/A04_R1_acls_ospf.png
Figure 5	R3-DC – DNS server records for corp.local and OSPF database summary	screenshots/A05_R3_dns_server.png
Figure 6	R2-BR – Branch sub-interfaces, DHCP bindings, OSPF adjacency and inter-area routes	screenshots/A06_R2_branch.png
Figure 7	SW1-HQ-DIST – VLAN database and 802.1Q trunks	screenshots/A07_SW1_vlans_trunks.png
Figure 8	SW2-HQ-ACC – spanning tree: redundant uplink Fa1/2 in BLK state (loop-free redundancy)	screenshots/A08_SW2_stp_ports.png
Figure 9	R1-HQ – SSH v2, NTP synchronisation to R4-CORE, remote Syslog and VTY hardening	screenshots/A09_R1_ssh_ntp_syslog.png
Figure 10	PC-IT1 – DHCP lease, DNS resolution of web.corp.local, inter-site and Internet reachability, traceroute via the core	screenshots/A10_PC-IT1_dhcp_dns_connectivity.png
Figure 11	PC-GUEST1 – Internet and DNS allowed; internal servers and Finance blocked by GUEST-IN	screenshots/A11_PC-GUEST1_acl_internet_only.png
Figure 12	PC-SALES1 – web server (HTTP) reachable, Finance VLAN blocked by FINANCE-PROTECT	screenshots/A12_PC-SALES1_web_finance_blocked.png
Figure 13	R1-HQ – ACL hit counters after the tests prove the policy is enforced	screenshots/A13_R1_acl_hit_counters.png
Figure 14	ANSIBLE-SRV – central Syslog collector receiving messages from all network devices	screenshots/A14_syslog_collector.png
Figure 15	Ansible project tree, version and inventory graph on the automation server	screenshots/B01_ansible_version_inventory.png
Figure 16	Ansible Vault – credentials encrypted with AES-256; only variable names are ever displayed	screenshots/B02_ansible_vault.png
Figure 17	00_facts_inventory.yml – ios_facts discovery of all 10 devices and the generated inventory CSV	screenshots/B03_facts_inventory.png
Figure 18	01_backup.yml – timestamped running-configuration backups plus a 'latest' copy per device for Git diffing	screenshots/B04_backup.png
Figure 19	02_baseline.yml – hardening, SSH, NTP, Syslog, DNS, archive logging, SNMPv3 and banner enforced on all devices	screenshots/B05_baseline.png

Figure	Caption	File
Figure 20	03_switching.yml – VLAN database, 802.1Q trunks, access ports and STP root managed on the six switches	screenshots/B06_switching.png
Figure 21	04_routing.yml – Jinja2-rendered sub-interfaces, P2P links and multi-area OSPF match the running network (changed=0)	screenshots/B07_routing.png
Figure 22	05_services.yml – DHCP pools, DNS host records and NAT/PAT rendered from site data	screenshots/B08_services.png
Figure 23	06_security_acl.yml – declarative ACLs (whole block replaced only if the device differs) and live hit counters	screenshots/B09_acl.png
Figure 24	07_verify.yml – automated assertions: OSPF FULL, default route, VLANs, trunks and reachability on every device	screenshots/B10_verify.png
Figure 25	Verification report rendered by the verify role (reports/verification_report.md)	screenshots/B11_verify_report.png
Figure 26	08_compliance.yml – 14 security rules audited on 10 devices: 14/14 on every device	screenshots/B12_compliance.png
Figure 27	Idempotency – the complete configuration pipeline (playbooks 02–06) re-run: changed=0, failed=0 on all 10 devices	screenshots/B13_idempotency.png
Figure 28	CHG-0001 dry run (--check --diff): Ansible predicts the changes (VLAN 45 would be created) without touching any device	screenshots/B14_day2_check_diff.png
Figure 29	CHG-0001 applied to the Branch devices (SW3-BR-DIST, SW4-BR-ACC, R2-BR) – failed=0	screenshots/B15_day2_apply.png
Figure 30	CHG-0001 verified on R2-BR: Fa0/0.45 gateway 10.20.45.1, DHCP pool BR-MARKETING, OSPF area 20	screenshots/B16_day2_verify_R2.png
Figure 31	CHG-0001 verified on SW4-BR-ACC: VLAN 45 MARKETING present in the VLAN database and carried on the trunks	screenshots/B17_day2_verify_SW4.png
Figure 32	SSH v2 management by Ansible: legal banner from the baseline role and the live netadmin SSH session (aes128-cbc)	screenshots/B18_ssh_banner.png
Figure 33	AWS architecture provisioned by Terraform (ap-southeast-2)	docs/aws_architecture.png
Figure 34	terraform init / fmt / validate	screenshots/C01_terraform_init_validate.png
Figure 35	terraform plan – 25 resources to add, with outputs	screenshots/C02_terraform_plan.png
Figure 36	Git commit history (git log --graph) demonstrating version control of the whole project	screenshots/C03_git_history.png
Figure 37	git show of the Day-2 change – a four-line YAML diff drives a multi-device network change	screenshots/C04_git_day2_change.png